



Solar Wind and Pickup Ion (SWAPI) Instrument on NASA's Interstellar Mapping and Acceleration Probe (IMAP)

J.S. Rankin¹ · D.J. McComas¹ · M. Alimaganbetov¹ · N. Angold¹ · G.F. Dunn² · H.A. Elliott² · D. Everett¹ · J.D. Escobar¹ · M.B. Galvin¹ · L.Y. Khoo¹ · J.T. Letzer¹ · E.M. Roemer¹ · B. Savage¹ · M. Shaw-Lecrerf¹ · M.M. Shen¹ · B.L. Shrestha¹ · J. Teifert¹ · S.E. Weidner¹ · E.J. Zirnstein¹ · E.R. Christian³ · M. Gkioulidou⁴ · G. Nicolaou⁵ · N.A. Schwadron^{6,1} · P. Swaczyna⁷ · M.B. Tapley²

Received: 2 September 2025 / Accepted: 11 October 2025 / Published online: 11 November 2025
© The Author(s) 2025

Abstract

The Solar Wind and Pickup Ion (SWAPI) instrument onboard the Interstellar Mapping and Acceleration Probe (IMAP) is a top-hat electrostatic analyzer designed to measure energy-per-charge distributions of solar wind protons (H^+), alpha particles (He^{2+}), and interstellar pickup ions (PUIs; combined H^+ and He^+) across a range of 0.1 to 20 keV/q. These measurements are essential for advancing understanding of the solar wind dynamics, particle acceleration, and physical processes governing the heliosphere. SWAPI builds on the heritage of the Solar Wind Around Pluto (SWAP) instrument on New Horizons, with enhancements tailored for continuous operation at 1 au. A key innovation is its grounded aperture grid assembly, which passively attenuates solar wind flux by approximately three orders of magnitude while preserving a large geometric factor for PUIs. The instrument's electro-optics include an electrostatic analyzer, a field-free flight path, and a coincidence detector system employing an ultrathin carbon foil and dual Channel Electron Multipliers (CEMs), enabling effective background suppression with some species discrimination. These capabilities support detailed studies of solar wind transients, pickup ion distributions, and the interaction between the solar wind and the interstellar medium. SWAPI also contributes to IMAP's space weather system, extending ACE-like solar wind monitoring with enhanced time and energy resolution, and provides real-time data for heliospheric modeling and forecasting.

✉ J.S. Rankin
jsrankin@princeton.edu

¹ Department of Astrophysical Sciences, Princeton University, Princeton, NJ 08544, USA

² Southwest Research Institute, San Antonio, TX 78228, USA

³ Heliophysics Science Division, NASA Goddard Space Flight Center, Greenbelt, MD 20771, USA

⁴ Applied Physics Laboratory, Johns Hopkins University, Laurel, MD 20723, USA

⁵ Department of Space and Climate Physics, Mullard Space Science Laboratory, University College London, Dorking, Surrey, RH5 6NT, UK

⁶ Space Science Center, University of New Hampshire, Durham, NH 03824, USA

⁷ Space Research Centre, PAS (CBK PAN), Bartycka 18A, Warsaw, Poland

Keywords Plasma · Solar wind · Pickup ions · Solar activity · Heliophysics · Space weather

1 Introduction

The IMAP mission has four principal science objectives, as described in (McComas et al. 2018, 2025b, *italics*): 1) *improve understanding of the composition and properties of the local interstellar medium (LISM)*; 2) *advance understanding of the temporal and spatial evolution of the boundary region in which the solar wind and the interstellar medium interact*; 3) *identify and advance the understanding of processes related to the interactions of the magnetic field of the Sun and the LISM*; and 4) *identify and advance understanding of particle injection and acceleration processes near the Sun, in the heliosphere and heliosheath*.

The Solar Wind and Pickup Ion (SWAPI) instrument onboard IMAP is a next-generation sensor derived from the Solar Wind Around Pluto (SWAP) instrument (McComas et al. 2008) flown on the NASA New Horizons planetary mission (Stern 2008) that flew past Pluto and is currently operating in the outer heliosphere, beyond 60 au. SWAPI primarily addresses IMAP's Science Objectives 1 & 4, contributing to a deeper understanding of the solar wind (SW) environment, the sources and acceleration of particles, the behavior of interstellar pickup ions (PUIs) and the physical processes shaping the global heliosphere (see Table 1). In support of Objective 1, SWAPI provides dedicated measurements of the He^+ PUI distribution across a broad energy range (~ 0.1 to 20 keV/q) at a 10 min cadence for the first time. These high time-resolution observations reveal the detailed structure of the Helium PUI focusing cone (observed by SWAPI based on interstellar pickup He^+ measurements), providing a unique means to probe the LISM. When combined with complementary measurements from the Compact Dual Ion Composition Experiment (CoDICE; Livi et al. 2025), IMAP-Lo (Schwadron et al. 2025), and Global Solar Wind Structure (GLOWS; Bzowski et al. 2025), these data enable determination of the LISM flow properties with unprecedented accuracy (Cohen et al. 2025; Szalay et al. 2025). For Objective 4, SWAPI delivers the time and energy resolution necessary to resolve spatial structures and transients in the SW, such as Coronal Mass Ejections (CMEs), as well as the ultra-fast (≤ 1 min) PUI observations to study the PUI pickup and acceleration processes. These observations are essential for identifying energetic particle sources and linking these to local acceleration processes (Cohen et al. 2025). SWAPI's measurements include plasma properties derived from 1D velocity distribution functions (VDFs) of SW protons (H^+) and alphas ($^4\text{He}^{++}$), as well as the first high-cadence observations of the suprathermal tail and PUIs at 1 au.

SWAPI also provides continuous real-time SW data for the IMAP Active Link for Real-Time (I-ALiRT) system similar to prior real-time measurements on ACE, STEREO, Wind, and DSCOVR, but with higher time resolution and new capabilities (McComas et al. 2025b; Lee et al. 2025). Specifically, SWAPI extends the ACE measurements of the SW plasma by providing high-time resolution (< 15 s required; exceeds at 12 s) measurements of the

Table 1 SWAPI contributions to IMAP science objectives

Objective	Science Target	Measurement
LISM flow properties (1)	Interstellar inflow longitude direction from the He^+ PUI focusing cone	PUI He^+ 1D VDFs
Origin of particle acceleration throughout the heliosphere (4)	High-resolution (< 1 min) suprathermal tails, and accelerated particles	Solar Wind H^+ & $^4\text{He}^{++}$ distribution functions

Table 2 SWAPI instrument performance. *“GE” refers to the effective geometric factor, which includes the efficiency factor (i.e., εG as opposed to G , where ε refers to the instrument efficiency and G is the geometric factor)

Parameter	Performance
Instrument Type	Top-hat top plate Electrostatic Analyzer
Species	SW H^+ & $^4He^{++}$; combined PUI He^+ & H^+
Energy Range	0.089 to 21.4 keV/q
Energy Resolution ($\Delta E/E$)	8.1% Full Width at Half Maximum (FWHM)
Time Resolution	12 s
Number of Steps	72 (1 ramp, 62 coarse, 9 fine)
Field of View	Sun-viewing: $30^\circ \times 10^\circ$ Non Sun-Viewing: $221^\circ \times 20^\circ$ (factors in vignetting)
GE ($cm^2 \text{ sr eV/eV}$)*	SW Sun-Viewing: $1.00 \times 10^{-7} \text{ cm}^2 \text{ sr eV/eV}$ for SW H^+ (cold plasma w/ 5° FWHM beam) PUI Non-Sun-Viewing: $2.25 \times 10^{-2} \text{ cm}^2 \text{ sr eV/eV}$ for PUI He^+ (hot plasma approximation)
Dynamic Range	2 MHz to 0.07 Hz counting rates
Signal to Noise Ratio	Min. SNR of 1 @ 0.07 Hz coincident count rate
Mass	3.01 kg
Volume	0.11 m^3
Power	2.37 W (peak power, 12 s cycle average)
Telemetry	443.2 bps (housekeeping + science)

SW plasma 1D VDF (0.4 to 10 keV/q in energy) with an energy resolution of 8.1% and 27 energy bands per decade.

2 Science Requirements & Capabilities

SWAPI’s science requirements focus on the three main particle populations measured in situ at 1 au: Solar Wind protons (SW H^+), Solar Wind alphas (SW $^4He^{++}$) and the combined interstellar pickup ion (PUI) distribution of protons and helium (PUI H^+ & He^+), which is dominated by helium at 1 au (see, e.g., Zirnstein et al. 2022). These are summarized in Table 12 of Appendix A, with key performance parameters provided in Table 2. In all cases, SWAPI has been designed, tested, and demonstrated to meet, and in most cases exceed, requirements that have been specified since the beginning of the IMAP Mission (McComas et al. 2018).

3 High Level Sensor Description

SWAPI is a top-hat electrostatic analyzer designed to make combined measurements of solar wind (SW) and pickup ions (PUIs) at 1 au (Fig. 1). Building on heritage from SWAP (McComas et al. 2008), SWAPI incorporates several key optimizations for operations on IMAP at 1 au. These include an expanded energy range of 0.1 to 20 keV/q (bin centers; compared to SWAP’s 0.04 to 7.5 keV/q) and wider-angle field-of-views (FOVs), enabled by the removal of the SWAP Retarding Potential Analyzer (RPA) and deflector ring. The

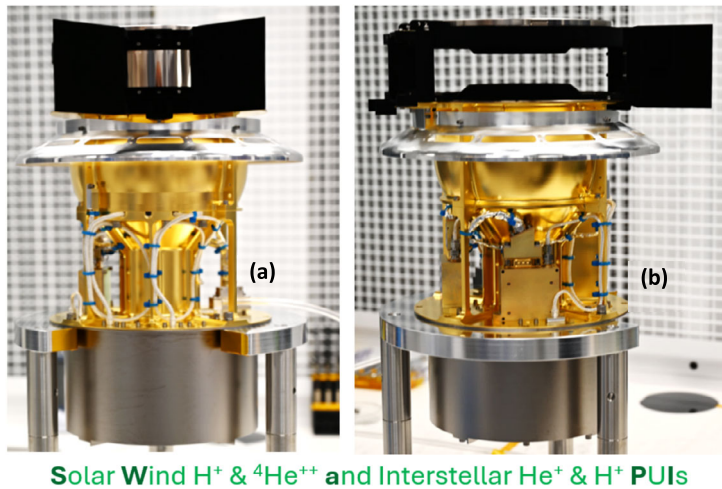


Fig. 1 The Solar Wind and Pickup Ion (SWAPI) instrument onboard IMAP is designed to simultaneously measure Solar Wind (H^+ and $^4He^{++}$) and Pickup Ions (combined H^+ and He^+) at 1 min and 10 min time resolutions, respectively. Panel (a) shows the instrument's solar-wind-facing FOV, complete with the $\sim 0.1\%$ transmissive grid ("sunglasses") and vanes, while panel (b) shows the unrestricted open aperture FOV for pickup ions (rotated 90° with respect to the solar wind configuration)

electro-optics include a grounded aperture grid assembly with vanes and "sunglasses", an aperture cutoff ring that restricts the elevation-angle FOV to $\pm 10^\circ$ in order to avoid direct line-of-sight with the spacecraft, a grounded Outer ESA Dome, a variable-voltage Inner ESA Dome, a field-free flight path, and a detector section (Fig. 2).

The simultaneous observations of vastly different particle populations is enabled by the entrance aperture via a SW-viewing aperture grid assembly (Figs. 1a & 2, left) – comprised of a grid to attenuate the nominal SW and vanes to block any unattenuated SW – and an open aperture with a wide FOV for interstellar PUIs (Figs. 1b & 2, right). Particles are selected based on their energy per charge (E/q) using voltages applied to the Inner ESA Dome, while the Outer ESA Dome is held at ground. Those within an appropriate range of energies and angles (e.g., within the energy-angle passband) traverse the entrance aperture and are post-accelerated into the detector section.

The detector section consists of two channel electron multipliers (CEMs; the Primary, or "PCEM", and Secondary, or "SCEM") separated by an ultra-thin carbon foil. Ions that pass through the foil are detected by the PCEM, while electrons that scatter off the front of the foil are detected by the SCEM. Each particle impact on a CEM generates a cascade of secondary electrons, forming a charge pulse at the input of a charge amplifier (CHAMP). Signals from the primary and secondary CEMs are amplified and shaped by the CHAMPs before being converted into logic-level pulses, which accumulate in digital counters implemented on SWAPI's Control Board. If primary and secondary counts are registered within the same 99 ns window, a coincidence (COIN) count is also recorded.

A photo capturing key as-built components of the SWAPI flight instrument is shown in Fig. 3.

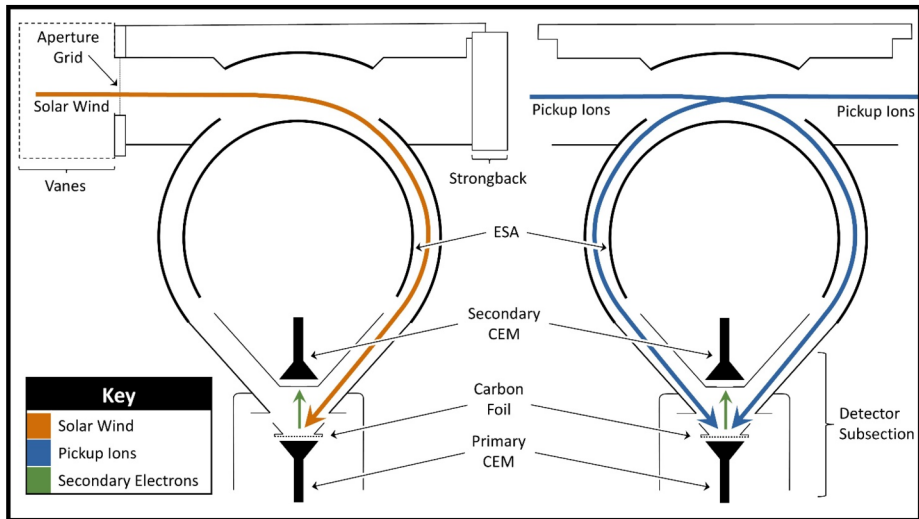


Fig. 2 Cross-sectional view of the SWAPI electro-optics, shown for the SW (right) and PUI (left) trajectories. Orange trajectories represent incoming SW ions, while blue highlights the PUIs. Ions of selectable energies and angles travel between the Inner and Outer ESA Domes through a field-free flight path region and eventually pass through an ultra-thin carbon foil into the Primary CEM. Green trajectories show examples of secondary electrons emitted from the front side of the foil and received by the Secondary CEM to provide the coincidence timing

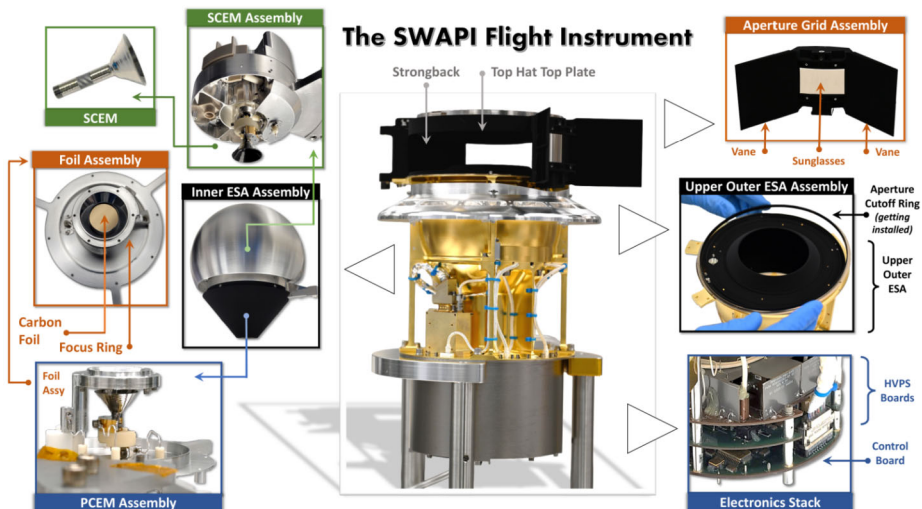
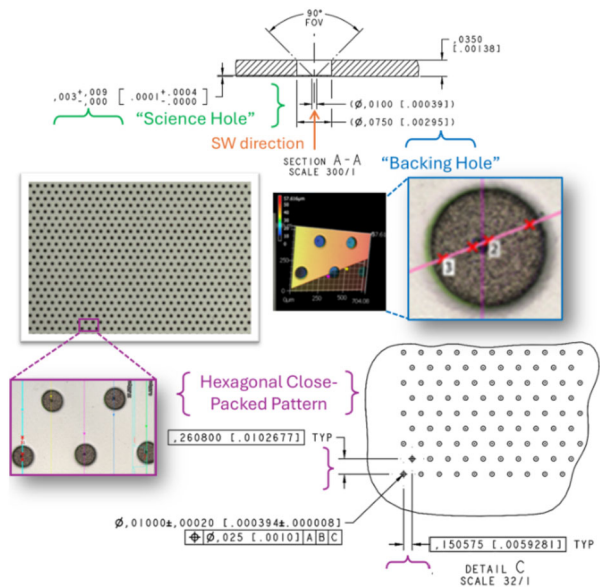


Fig. 3 Photo of the fully integrated SWAPI flight instrument, with key components highlighted

3.1 Entrance Aperture Design

As mentioned above, the aperture grid assembly consists of two primary components: a 30°-wide, thin electroformed nickel grid designed to attenuate the high-intensity SW – nicknamed the “sunglasses” – and a set of protruding vanes to manage particle vignetting. This

Fig. 4 Mechanical drawings and inspection photos of SWAPI's aperture grid holes at varying levels of magnification



configuration ensures that 99.9% of the SW is blocked ($\sim 0.1\%$ transmission) 99.99% of the time (30° -wide SW-viewing aperture) while the angle of the vanes ($\sim 40^\circ$ with respect to the SW-pointed direction) ensures that the nominal SW is prevented from scattering within the electrostatic analyzer 99% of the time (Fig. 1a). The remaining $\sim 221^\circ$ open-aperture region allows unrestricted entrance of PUIs, maximizing the geometric factor for these much weaker signals (Fig. 1b).

3.1.1 Solar Wind Attenuating Aperture Grid (“Sunglasses”)

The dual-layer electroformed nickel grid was designed to uniformly attenuate the SW while minimizing particle scattering and maintaining thermal and mechanical stability. The front layer is ~ 9 to $12\ \mu\text{m}$ thick with 21,460 precision holes averaging $10.325\ \mu\text{m}$ in diameter (“science holes”), while the backing layer is $\sim 26\ \mu\text{m}$ thick with oversized holes ($\sim 103\ \mu\text{m}$) aligned to the front layer (“backing holes”). The grid covers a $52.7\ \text{mm} \times 32.3\ \text{mm}$ area with holes arranged in a hexagonally close-packed pattern. Its overall thickness is $\sim 35\ \mu\text{m}$. This design evolved from a single-layer concept after thermal modeling revealed excessive heating ($>230\ ^\circ\text{C}$) and risk of distortion. The dual-layer configuration reduces peak temperatures to $\sim 90\ ^\circ\text{C}$, ensuring mechanical integrity even under the constant solar exposure. The flight grid was selected after microscopic and laser inspection at Princeton University’s Micro/Nano Fabrication Laboratory, and calibration confirmed that this novel grid design achieves the intended uniform attenuation of $\sim 0.1\%$ (Sect. 4.1.2, Fig. 18). Drawings of the holes, along with images from inspection, are shown in Fig. 4.

3.2 Electrostatic Analyzer and Particle Selection

During nominal operations (“HVSCI mode”), SWAPI sweeps the ESA voltage from $-10,240\ \text{V}$ to $-50\ \text{V}$, which, with an energy resolution of 8.1% corresponds to a full energy range of $21,364\ \text{eV}$ to $89\ \text{eV}$ for an analyzer constant of $k = 1.93$. This sweep occurs

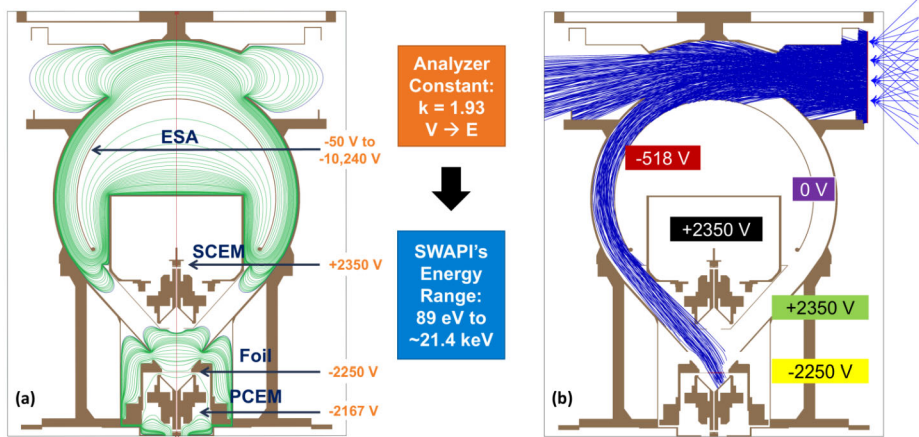
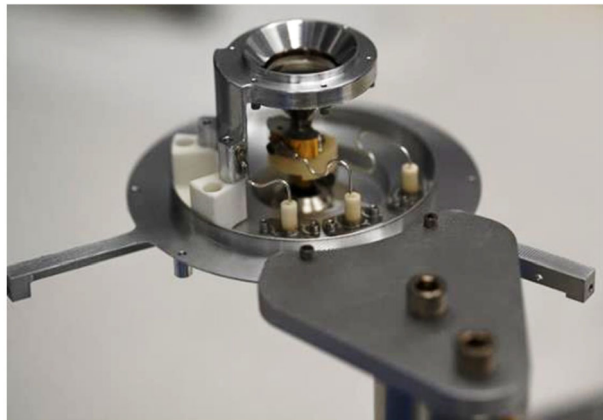


Fig. 5 Cutaways of SWAPI showing nominal fixed voltages on the PCEM, SCEM, and foil and the range of voltages applied to the ESA, in the form of equipotentials (panel a) and trajectories of 1 keV protons at a variety of incident angles (panel b) generated by the instrument electro-optical model (SIMION[®]). For an analyzer constant of $k = 1.93$, the optimal ESA voltage to detect the 1 keV particles is -518 V

Fig. 6 PCEM and focus ring assembly that holds the ultra-thin carbon foil



every 12 s using a 72-step sequence defined in the flight lookup table (LUT; 26850.06-LUT-01.bin), which includes one ramp-up step, 62 steps across the full energy range, and 9 variable steps centered on the peak E/q bin. Particle trajectories and electrostatic equipotentials derived SIMION[®] modeling are shown in Fig. 5.

3.3 Detector Subsystem

The detector subsystem is optimized to provide coincidence detection of incoming ions. After passing through the ESA and field-free region (e.g., Fig. 5a), ions are accelerated by high CEM voltages toward a focus ring that holds a nominally $1 \mu\text{g}/\text{cm}^2$ thick ultra-thin carbon foil (McComas et al. 2004) suspended on a 65% transmissive nickel grid (Fig. 6). In normal operations, the focus ring and foil are biased by the PCEM HV supply output, with a resistive divider maintaining a ~ 100 V potential difference between the focus ring

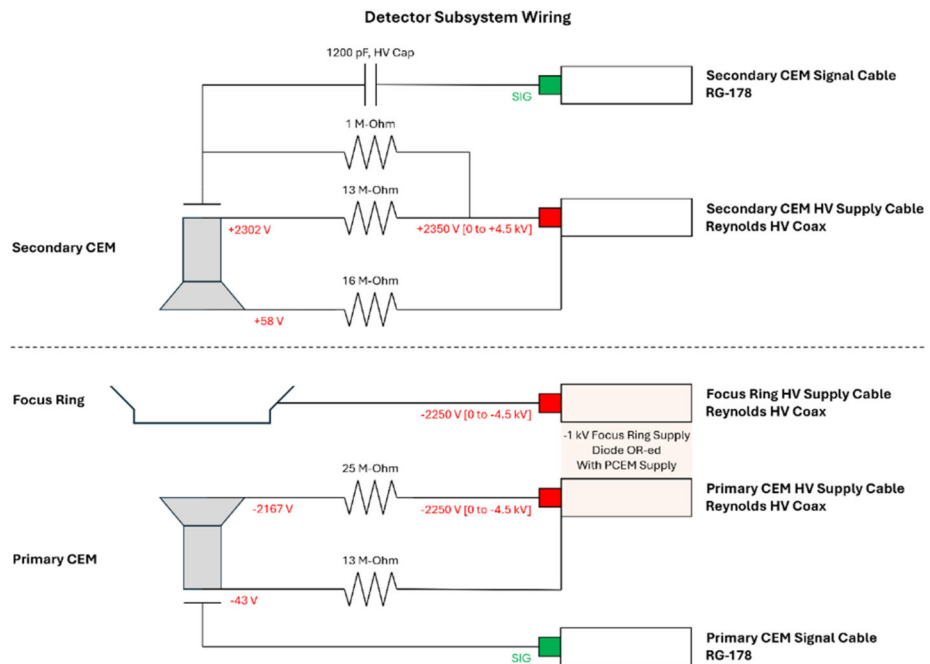


Fig. 7 SWAPI Detector subsystem wiring

and primary detector. This configuration accelerates forward-scattered electrons into the PCEM and directs backward-scattered electrons to the SCEM. Either signal can initiate the 99-ns coincidence timer. SWAPI's front end electronics counts the discrete avalanche events from each of the CEM detectors and can resolve individual detector pulses at rates up to $\sim 5\times$ higher than the expected maximum count rates, ensuring accurate event detection without ambiguity (see Sect. 3.5). The dual-detector design, inherited from SWAP, provides redundancy and enables continued operation even if one detector is disabled (see Fig. 7 for wiring diagram).

Redundant power for the focus ring and foil is provided by a dedicated multiplier chain, diode-OR-ed with the PCEM supply output (highlighted by the pink box in Fig. 7). If the PCEM supply is disabled, the focus ring multiplier can deliver a bias voltage between 0 and -1 kV, loosely controlled by the drive signal it shares with the SCEM HV supply. To preserve detector longevity, only ultra-low outgassing materials are used near the CEMs, following practices established in earlier missions (e.g., McComas et al. 1987).

3.3.1 Carbon Foil

The ultra-thin carbon foil was floated at the Princeton Space Physics Laboratory. It consists of a nominal ~ 5 nm ($1.0 \mu\text{g}/\text{cm}^2$) layer of amorphous carbon suspended on a 333 line-per-inch nickel grid. The foil was inspected using a Nikon camera with a macro lens and a digital microscope, and image analysis confirmed the flight foil's excellent coverage ($\sim 99.8\%$) with minimal blown-out holes. A dedicated acoustic test was performed to verify mechanical integrity under launch conditions.

3.3.2 Channel Electron Multipliers (CEMs)

SWAPI's quad-spiral CEMs have a resistance of approximately 600 M Ω and dark count rates below 0.04 s⁻¹. The coincidence detection system significantly reduces background noise from intrinsic detector activity, UV photons, cosmic rays, and other high-energy particles. Flight-qualified CEMs were selected through a rigorous screening process. Early operational gain variability due to outgassing from channel surfaces was mitigated through UV-induced burn-in, where controlled UV exposure conditions the detectors until gain stabilizes (McComas and Bame 1984). After burn-in, detectors were characterized for resistance, gain and count rate versus applied voltage, and dark counts using a custom system in the Princeton Space Physics laboratory (see Appendix B). Once integrated into SWAPI, gain curve tests were performed to tune the CEM bias voltages, ensuring that secondary electrons from the carbon foil produce pulses above the amplifier's discriminator threshold. These tests are repeated on-orbit to compensate for gain degradation over time (see Sect. 4.3).

3.4 Mechanical Design

SWAPI's mechanical design builds on the SWAP architecture, with modifications to accommodate higher-energy measurements and mission-specific constraints. The instrument is divided into two main subassemblies: the optics, which guide particles to the detectors (described above) and the instrument electronics (described below), which include the control board, High Voltage Power Supply (HVPS), two charge amplifiers (CHAMPs), and associated cabling. Structural and thermal design considerations were incorporated to meet flight requirements.

3.4.1 Optics & Detector Assembly

Electro-optical changes from the heritage design include removal of SWAP's RPA and deflector, and the addition of the sunglasses, vanes, and aperture cutoff ring. The removal of deployable doors and actuators leaves a large section of the aperture wide open, which necessitated the design and manufacture of a custom cleanroom-class cloth red tag cover to reduce nitrogen purge leak rates. Internal surfaces were coated with Ebonol-C, including the aperture region and components such as the top hat and heritage door hinge support (changed from the electroless nickel coating on SWAP). Although the Inner ESA Dome was also originally intended to be coated in Ebonol-C, late-stage schedule constraints necessitated the adoption of microgrooves. This, in conjunction with the Upper and Lower Outer ESA Ebonol-C surfaces and serrated grooves proved effective for meeting instrument requirements, as validated through rigorous UV testing (see Sect. 4.1.2, Table 6). The Inner ESA Dome was also redesigned with an 0.04" diameter teardrop bottom edge and rounded internal geometry to support higher voltages.

Higher operating voltages required additional design modifications near the detector assemblies. Alumina insulators used in SWAP were replaced with larger PEEK insulators to isolate high voltage regions in the vicinity of the Inner ESA Dome from ground. Polished surfaces were introduced around the detectors and ESA to prevent arcing, and the ESA feedthrough design was updated to accommodate the increased voltage for the connection that powers the Inner ESA Dome (Fig. 8).

To meet IMAP's launch requirements, a custom stitched blanket of MLI held in place with conductive aluminum adhesive tape was added to seal gaps between the MLI tie point and the Upper Outer ESA Dome, preventing trapped gas from venting directly toward the sensitive aperture.

Fig. 8 Partially assembled Inner ESA Dome assembly, including the “teardrop” edge and micro-groove finish of the Dome, PEEK insulators, the SCEM, its HV feedthroughs, and rounded/polished edges adopted throughout the HV regions of the assembly

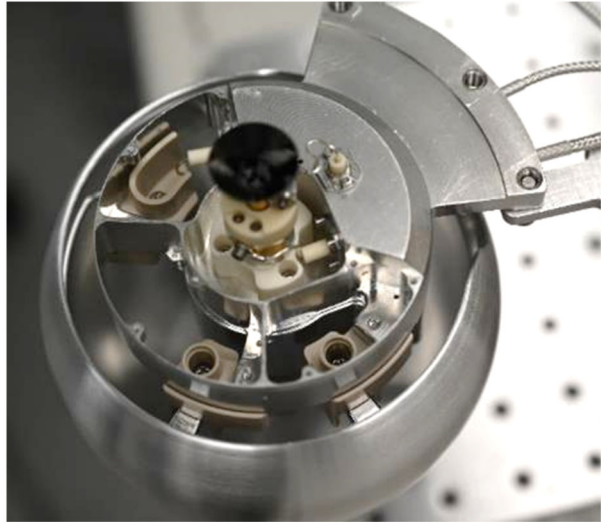


Fig. 9 Assorted perspectives of SWAPI’s HV cables, highlighting silver-plated copper braids, and rounded solder “donuts”



3.4.2 Electronics Packaging and Cabling Design

The SWAPI electronics subassembly consists of three vertically stacked circuit boards housed within a dedicated enclosure. These include: (1) the Control Board with Low Voltage Power Supplies (LVPS) on the bottom; and two High Voltage Power Supply (HVPS) boards: (2) the Driver Board (middle) and (3) the HV Base Board (top), which includes voltage multiplier assemblies for the ESA, PCEM, SCCEM, and Focus Ring. Electrical connections between boards are made via discrete pins and sockets. Standoffs provide both mechanical support and facilitate thermal conduction to the instrument’s chassis. The electronics subassembly is covered in more detail in Sect. 3.5.

Each CEM signal is first processed by a dedicated CHAMP module (see Sect. 3.5.1) before entering the electronics box. The CHAMP modules are mounted externally on the electro-optics and detector bulkhead to minimize noise by distancing them from digital electronics.

High voltage cables are soldered directly to the HV Base Board and are strain-relieved using machined PEEK blocks. Terminations at the detectors and electro-optics are made via ring terminals. To mitigate arcing risks, cable shields are terminated using smooth HV termination “donuts” (Fig. 9). Where exposed outside the shielded electronics enclosure,

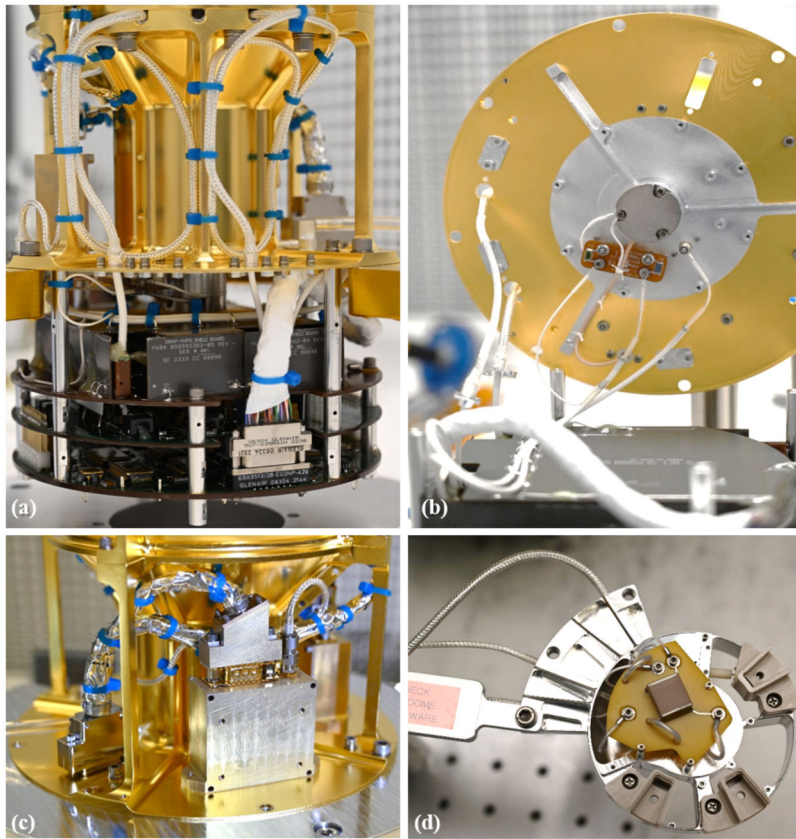


Fig. 10 As-built SWAPI electronics and cable packaging. (a) Internal view of the electronics enclosure, showing the board stack and cable routing (HVPS Multiplier and Driver Boards on top and in the middle; Control Board on bottom). (b) Termination of PCEM power and signal cables underneath the Lower Outer ESA bulkhead. (c) CHAMP modules mounted externally to the sensor bulkhead. (d) Termination of SCEM power and signal cables within the Inner ESA Dome assembly

coaxial cables are sleeved with an additional layer of silver-plated copper braid, terminated to the instrument chassis.

All of the electronics are packaged in volumes isolated from the detectors and electro-optics to prevent contamination. The CHAMP modules, though external to the primary electronics enclosure, are enclosed in shielded housings. The wall thickness of both the main electronics enclosure and CHAMP housing meets mission requirements for Total Ionizing Dose (TID) protection.

Figure 10 illustrates the as-built configuration for the electronics packaging and cable routing described above.

3.4.3 Structural Design

The IMAP Component Mechanical Environmental Requirements Document (CMERD) defined the following critical structural requirements for SWAPI: (1) quasi-static load of 50g force applied separately along three orthogonal axes, (2) first mode structural fre-

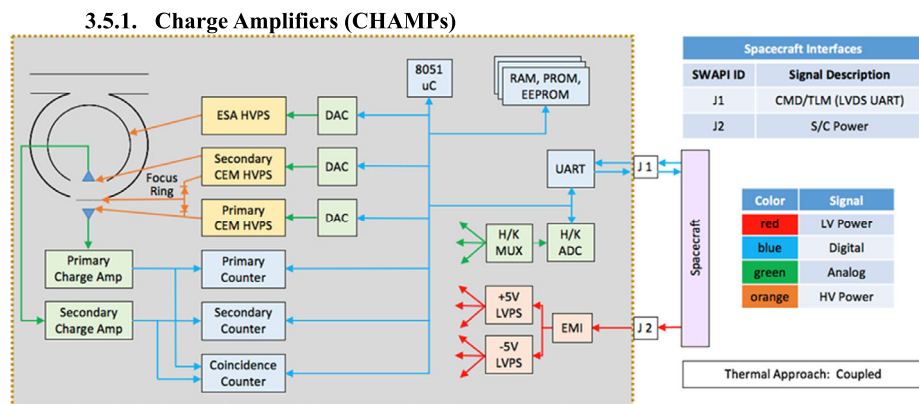


Fig. 11 SWAPI electronics block diagram

quency > 100 Hz (3) sine vibration up to 22g (30–34 Hz, in the axis perpendicular to the SWAPI bracket's mounting interface), and (4) random vibration with an overall amplitude of 12.8 Grms.

A structural analysis was conducted during the design phase to ensure the SWAPI instrument met these requirements. The analysis focused on structurally critical components, including the cantilevered Inner ESA Dome and field shaper, as well as the minimally supported top hat and aperture grid subassembly. Results indicated multiple candidate first-mode resonances. In all cases, predicted modal frequencies well-exceeded 100 Hz. Vibration testing of both the Engineering Model (EM) and Flight Model (FM) verified that SWAPI instrument could withstand the prescribed levels. Following the EM vibe tests, extensive multi-point microscopic inspections were also conducted to visually confirm structural integrity of locations analyzed to have margins.

3.4.4 Thermal Design

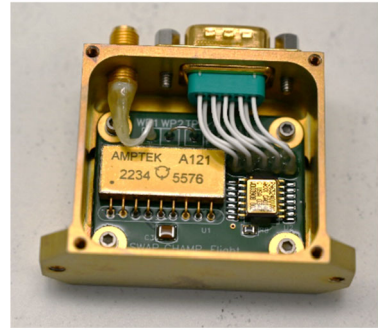
The CMERD defined the allowable temperature range at the SWAPI mounting interface as $-10\text{ }^{\circ}\text{C}$ to $+50\text{ }^{\circ}\text{C}$ for operation, and $-20\text{ }^{\circ}\text{C}$ to $+60\text{ }^{\circ}\text{C}$ for survival. Thermal analyses were conducted to ensure that all temperature-sensitive components – including electronics and detectors – would remain within their qualified limits under these conditions. The analysis also confirmed that the heat exchange with the spacecraft did not exceed 11 W, with the maximum occurring at the instrument's cold limit. To validate these results, a thermal vacuum test was performed, demonstrating full instrument functionality across the specified temperature range, including successful hot and cold turn-on sequences.

3.5 Electronics

A system-level block diagram of SWAPI's electronics is shown in Fig. 11.

Except for the electro-optics and detectors, all elements are integrated on printed wiring boards (PWBs) that were independently designed, fabricated, and tested. Interfaces between these elements were defined and managed at the system level to ensure seamless integration. Descriptions of each component's design and testing are provided in the following sections.

Fig. 12 Photograph of a SWAPI CHAMP



3.5.1 Charge Amplifiers (CHAMPs)

The SWAPI Charge Amplifiers are largely build-to-print replicas of those used on SWAP (see Sect. 3.4.1 of McComas et al. 2008), with one modification: a single resistor value was adjusted to reduce the CHAMP deadtime, enabling the higher count rates needed to satisfy SWAPI's science requirements.

The CHAMP electronics (shown in Fig. 12) convert low-current charge pulses from the CEMs into logic-level pulses, which are accumulated in digital counters on the Control Board. This is accomplished using a high-speed commercial hybrid device consisting of a charge-sensitive preamplifier and discriminator. The preamplifier outputs a voltage proportional to the integrated input current, while the discriminator generates a digital pulse when this voltage exceeds a configurable threshold known as the low-level discriminator (LLD) voltage. A 100-ohm resistor and back-to-back diodes protect the AC-coupled input from discharges.

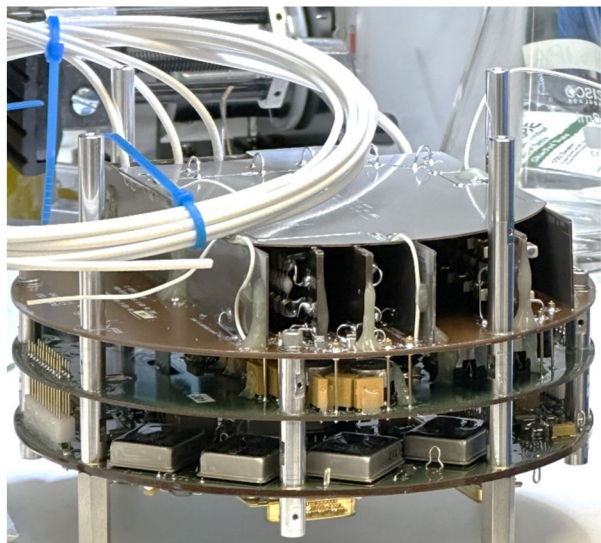
The Control Board interface is a 9-pin D-subminiature connector that provides power, LLD voltage threshold, test pulser input, and an event output signal. The LLD threshold voltage is set by a digital-to-analog converter (DAC) on the Control Board and can be updated via software commands. Output pulse width is set to 70 ns and deadtime to 230 ns using fixed resistors – supporting reliable performance at pulse rates exceeding 3 MHz. Output pulses are buffered through two Schmitt-trigger inverters with a series termination resistor at the driver output.

3.5.2 High Voltage Power Supply (HVPS)

The SWAPI HVPS is derived from the heritage design of the SWAP HVPS (see Sect. 3.4.2 of McComas et al. 2008), with several modifications to meet IMAP mission and SWAPI-specific requirements. Notably, the ESA HVPS output increased from -4 kV to -10.24 kV, and its settling time was reduced to 16.7 ms to support the required 6 energy steps per second cadence. The PCEM, SCEM, and Focus Ring retain the design and performance specification of the SWAP instrument, including the diode-OR configuration that ensures secondary electron acceleration for the SCEM detector regardless of whether the PCEM is turned on. A “zap trap” to chassis ground was added to the secondary ground to provide a controlled path for any discharge current. With the removal of the RPA and deflector from the electro-optic design, their associated HVPS units were also eliminated. Table 3 summarizes the key performance specifications for the SWAPI HVPS, and Fig. 13 shows the HVPS assembly integrated with the Control Board.

Table 3 Key Performance Requirements of the SWAPI HVPS

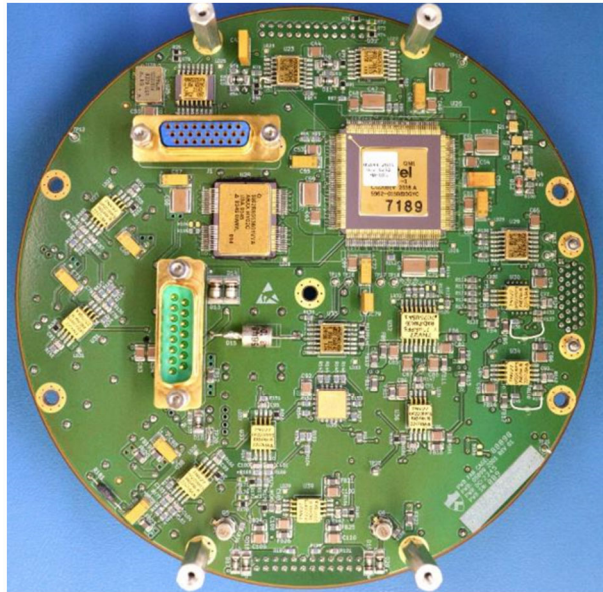
	PCEM	SCEM	Focus Ring	ESA Low Range	ESA High Range
Voltage Range [V]	0 to −4500	0 to +4500	0 to −1000	0 to −1024	0 to −10,240
Ripple [Vrms]	0.5	0.5	10	0.5	0.5
Large Scale Rise Time [ms]	n/a	n/a	n/a	0 to 10.24 kV $\leq$ 167 ms	
Large Scale Fall Time [ms]	n/a	n/a	n/a	10.24 kV to 51.2 V $\leq$ 183.3 ms (implies Tau-RC = 34.6 ms)	
Small Scale Rise Time [ms]	n/a	n/a	n/a	10% step to 1% accuracy In 16.7 ms	
Small Scale Fall Time [ms]	n/a	n/a	n/a	10% step to 1% accuracy In 16.7 ms	
Accuracy	± 5 V	± 5 V	$\pm 20\%$	$\pm 1\%$ for output > 51.2 V	$\pm 1\%$ for output > 512 V

Fig. 13 The SWAPI flight HVPS Assembly (top & middle) mated to the Control Board (bottom)

3.5.3 Control Board

SWAPI's Control Board, shown in Fig. 14, consolidates power distribution, data acquisition, high-voltage control, health and safety monitoring, CHAMP LLD threshold control, and the supply/conditioning of low-voltage power. The CHAMPs and external temperature transducer interface with the Control Board through the low-voltage instrument harness, connected via a 31-pin microminiature-D connector. Power regulation is achieved using four 1.5 W DC-DC converters that generate secondary ± 5 V power rails isolated from the spacecraft primary bus. One converter pair supplies the low-voltage electronics and CEM high-voltage supplies, while the other powers the ESA HVPS. An EMI filter conditions the input power and eliminates low-frequency beat noise from unsynchronized oscillators within the DC-DC converters. Secondary grounds are protected via “zap-traps” – back-to-back diodes that clamp transients to chassis ground. The primary power interface with the

Fig. 14 Photo of the SWAPI Control Board



spacecraft is a 15-pin D-subminiature connector with one conductor each for power supply and return.

Communication with the spacecraft is handled via a single non-redundant UART interface using LVDS signaling through a high-density D-subminiature connector with 26 sockets. Serial commands are received and telemetry transmitted over this interface at 115.2 kbaud. Instrument timing is synchronized using a virtual-pulse-per-second (PPS) signal embedded in a 1 Hz ITF frame from the spacecraft.

All onboard processing is performed by a radiation hardened 8051 microcontroller which handles command execution, high-voltage sequencing, data collection, and telemetry formatting. Operating at 4.9152 MHz with a minimum of 12 clock cycles per instruction, the processor achieves a maximum throughput of approximately 0.4 million instructions per second (MIPS). This architecture, inherited from SWAP (McComas et al. 2008), was retained to maximize code reuse and minimize power consumption.

SWAPI's memory architecture also mirrors that of its predecessor. It includes a 32k×8-bit PROM for Boot code, two redundant 128k x 8-bit EEPROMs for application code and LUTs (64 KiB each), and a 128k×8-bit SRAM divided into runtime storage areas for code (64 KiB each). Upon power-up, the boot code verifies EEPROM checksums and loads the valid application code into RAM for execution.

A field-programmable gate array (FPGA) manages access to all external memory and implements additional digital logic functions. These include control and status registers for instrument configuration, test pulsers for in-flight CHAMP and harness diagnostics, configurable timers for synchronizing data acquisition, digital counters for PCEM, SCEM, and coincidence events, DAC control for HVPS voltages and CHAMP LLD thresholds, and analog MUX and ADC control for health and safety monitoring. While the FPGA and CPU enable housekeeping telemetry for ground-based monitoring, SWAPI is designed to autonomously respond to critical out-of-limits conditions as described in Sect. 6.3.

To protect the CEMs from overcurrent conditions, the Control Board has dedicated hardware that continuously compares PCEM and SCEM supply currents against a configurable

analog threshold. If either current exceeds the threshold, a comparator circuit triggers a processor interrupt. The flight software interrupt service routine (ISR) responds by either reducing the commanded CEM voltage or disabling the supply, depending on the persistence of the overcurrent condition. Independent interrupt lines for the PCEM and SCEM allow for supply-specific responses.

3.5.4 Testing

Prior to integration, each SWAPI component underwent extensive testing to verify design integrity and workmanship. These tests included basic functional checks, detailed design verification, performance characterization over temperature and voltage extremes (i.e., four-corner testing), high voltage standoff qualification, and thermal cycle testing for both workmanship and environmental resilience.

Following successful validation at the component level, subsystem testing was conducted at each stage of integration. This included bench testing and calibration of the Control Board with the HVPS, as well as integration testing with the charge amplifiers.

3.6 Flight Software (FSW)

The SWAPI Flight Software (FSW) runs on the Control Board's 4.9152 MHz 80C51 micro-processor, which has a maximum instruction throughput of ~ 0.4 MIPs, as described above. The software is written primarily in C and is divided into two distinct software binaries: the Boot Flight Software (BootFSW) and the Application Flight Software (AppFSW). These components inherit significant heritage from SWAP.

BootFSW resides in a radiation-hardened, non-rewritable PROM, and executes immediately upon power-up. Its primary functions include low-level hardware initialization and self-tests, selection and boot of the AppFSW image, and entry into a failsafe/maintenance mode for diagnostics and memory operations if no valid AppFSW image is available. Once verified, the selected AppFSW is copied from EEPROM to SRAM for execution. AppFSW is stored in a reprogrammable EEPROM, with two redundant copies for fault tolerance. It provides the core operational functionality of the instrument, including fault detection and correction (FDC), HVPS control, memory management (upload/download), and science data collection and processing. Both software components synchronize with the spacecraft's 1 Hz Pulse Per Second (PPS) signal, which is subdivided into twenty 50-millisecond time slots to support deterministic task scheduling and alignment with spacecraft timing.

AppFSW outputs three types of telemetry: autonomy packets for onboard anomaly response engineering telemetry for housekeeping, event messages, and memory dumps, and science telemetry containing instrument data products. A summary of SWAPI's seven telemetry packets is provided in Table 4.

4 Calibration

SWAPI's calibration involves a combination of ground-based testing, cross-instrument comparisons, and in-flight operations to ensure accurate measurements throughout the mission. Instrument calibration established baseline measurements and verified compliance with science requirements. This was followed by a cross-calibration with the CoDICE instrument to align energy and angular measurements across overlapping ranges. On-orbit calibrations continue through periodic gain tests and routine data comparisons.

Table 4 SWAPI's telemetry packets

Packet	Mnemonic	Comment
Autonomy	SWP_AUT	Provided to the spacecraft at 1 Hz. Can be used to request power-down (e.g., power off/cycle flags). Not typically downlinked or written to the SSR.
Housekeeping	SWP_HK	Instrument health and status (mode, command counters, ADC values, etc.)
Memory Dump	SWP_MD	Packets of fixed size (128-byte payload), generated in response to a Memory Dump command.
Message	SWP_MG	Asynchronous information, warning, or error reports. Generated as needed.
I-ALiRT	SWP_IAL	Provides low-latency space weather data, transmitted to the spacecraft at 1 Hz while in HVSCI. Subset of Science packet (Coincidence Counts only). The spacecraft combines I-ALiRT data from all instruments into a single spacecraft I-ALiRT packet for downlink.
Science	SWP_SCI	Primary science data product. Each packet includes PCEM, SCEN, and Coincidence Counts for each of 6 samples acquired in a 1 s period. The ESA Level is also reported for a single sample; other levels can be determined from the LUT.
Large Science	SWP_LGSCI	Science packet containing detailed ESA level information for each HV step. Used for testing; same as SCI packet but includes all 6 ESA Levels.

Fig. 15 Photograph of the fully integrated SWAPI flight instrument, installed in the Princeton Main Chamber prior to calibration



4.1 Instrument Calibration

The calibration of the SWAPI flight instrument (referred to as the “flight calibration” from here on out) took place in the Princeton Space Physics Laboratory (Fig. 15; see Appendix B) and featured 40 tests performed over 15 days, achieving over 150 failure free operational

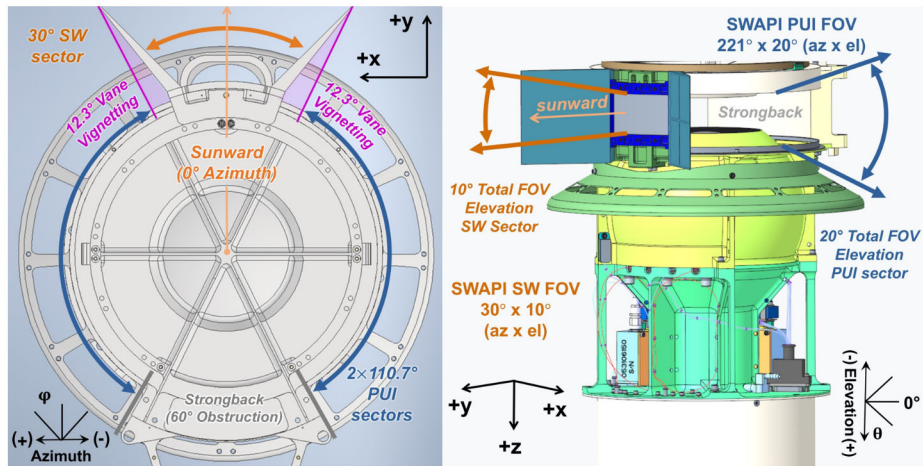


Fig. 16 Close-up perspectives of SWAPI's FOV in: (left) azimuth (ϕ) and (right) elevation (θ) angles, defined here with respect to the instrument coordinate system. In addition to the PUI and SW FOVs, there is a 60° region opposite to the SW sector that is obstructed by the “strongback”, a structure that supports the top hat top plate assembly (left panel); the 221° azimuthal FOV for the PUI sector (right panel) accounts for vignetting around both the strongback and the aperture grid vanes (see “PUI Open Aperture” description in Sect. 4.1.2)

hours. The calibration effort was designed to meet four primary objectives: (1) to characterize SWAPI's performance and verify compliance with science requirements; (2) to confirm functionality following environmental testing; (3) to measure key parameters for the response function and sensor model; and (4) to prepare for in-flight operations and on-orbit calibrations.

A major focus of the testing was to assess the uniformity of SWAPI's response over its azimuthal (ϕ) and elevation (θ) angle FOVs. The instrument's FOVs are defined as follows ($\Delta\phi \times \Delta\theta$), as illustrated in Fig. 16:

- SW-viewing: $30^\circ \times 10^\circ$
- Non-SW-viewing (PUI sectors): $221^\circ \times 20^\circ$.

4.1.1 Summary of Testing

A summary of testing is provided in Table 5. These tests were carried out with SWAPI mounted to a three-axis stage assembly that enabled linear translation and rotational motions in both azimuthal and elevation angles (see Fig. 16 and Appendix B, Figs. 41 & 42 for coordinate reference).

4.1.2 Instrument Performance Results

This section presents the results from selected examples of SWAPI's ground-based calibration tests outlined in Table 5. These demonstrate that the instrument meets its science requirements and also provide quantitative inputs for the sensor performance model and forward modeling efforts.

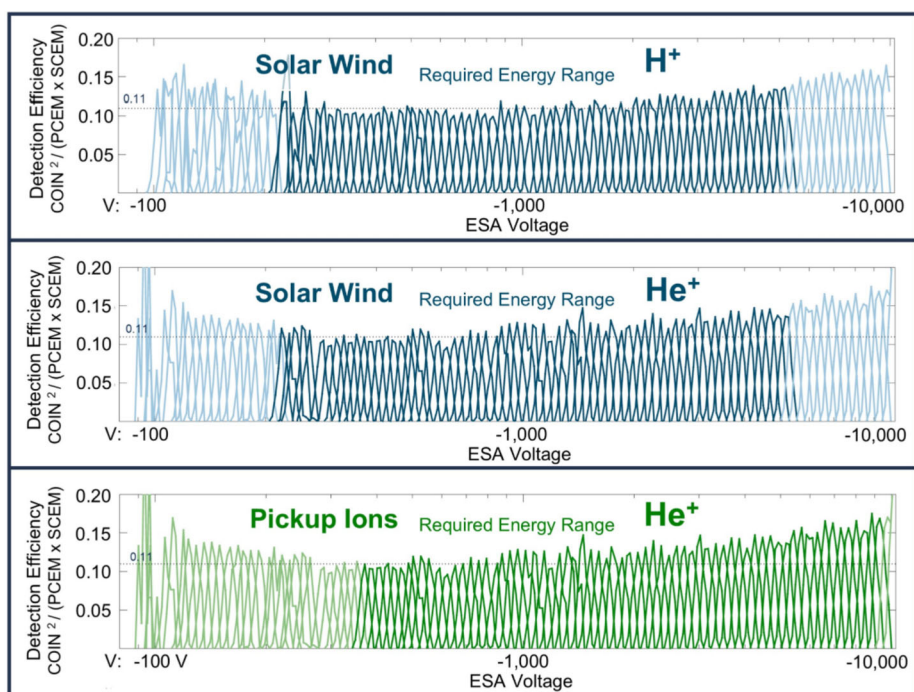
Table 5 Summary of tests performed during SWAPI's FM calibration campaign

Test Type	Description
Gain Curves	Used to determine optimal CEM voltages by sweeping one detector while holding the other fixed. Tests are performed both with static ESA voltages in engineering mode for full characterization and with flight-like ESA sweeps in science mode to simulate operational conditions. These tests are typically performed with stable ion beams at fixed instrument orientations.
Dark Counts	Performed to characterize intrinsic detector noise in the absence of beam or UV illumination, with the chamber isolated and CEM voltages set to their operational values. Count rates are additionally recorded over extended dwell periods to assess baseline noise at various ESA voltage settings in both engineering and science modes.
UV Background	A UV light source is introduced to evaluate susceptibility to stray light. CEMs remain fixed at nominal operating voltages while the instrument is exposed to UV radiation from a 1-Sun light source at various orientations. The test quantifies SWAPI's susceptibility to UV and identifies possible light-leak issues.
Continuous Energy Test	The instrument is parked at fixed azimuth and elevation angles while ESA voltages are stepped in engineering mode across a range of beam energies with sufficient resolution to map each beam. Ion source settings are adjusted to supply incrementing target energies. This test simultaneously characterizes detection efficiencies, energy resolutions and the instrumental k-factor over SWAPI's full range of operational voltages.
Energy-Angle Passband	Fixed at various locations in azimuth (e.g., to target SW and PUI sectors), the instrument is rotated through a range of elevation angles while ESA voltages are progressively stepped to map the energy-angle response. Multiple beam energies and species are used to define SWAPI's angular acceptance and energy resolution across its full elevation-angle FOV.
Angular Response Functions	Counts are accumulated while the instrument is positioned at various elevation and azimuthal angles, with a fixed-energy beam incident on SWAPI and the ESA set to an appropriate target voltage. Tests include azimuthal-elevation maps, which evaluate SWAPI's uniformity of detection over its angular FOVs, and azimuthal-angle rolls, which characterize the azimuthal response and sensitivity to structural features across the SW and PUI apertures. Tests are performed with SWAPI fixed at the relevant elevation angle and rotated in azimuth from $\pm 170^\circ$ and back at a continuous slow rate of 1° s^{-1} , and for one long test at an ultra-slow rate of $0.01^\circ \text{ s}^{-1}$ (1 keV H^+ to probe the sunglasses attenuation factor and aperture vignetting).

Characterization of Backgrounds and Signal Detection Thresholds Background tests such as dark counts and UV light tests were performed to set thresholds and characterize the instrument's signal-to-noise ratio. Tests were performed with CEMs at their operational voltages, chamber ion gauges turned off, and upstream gate valves closed to isolate the Main Chamber from the beam. Dark count rates were measured under two operational scenarios. In engineering mode (HVENG), the ESA was held at its minimum and maximum voltages for approximately 15 min each to evaluate detector behavior at fixed-energy settings. In science mode (HVSCI) the instrument performed flight-like voltage sweeps every 12 s over a 10-min interval, simulating nominal data collection across SWAPI's full instrument range. As summarized in Table 6, the individual count rates for the PCEM and SCEM remained below 0.1 Hz, while the coincidence rates consistently stayed below 0.01 Hz. These low background levels confirm the instrument's suitability for detecting the sparse PUI. UV background tests, also included in Table 6, revealed slightly elevated count rates on the individual CEMs compared to dark count conditions, as expected. However, no coincidence events were recorded during these tests, indicating that UV light does not produce false detections.

Table 6 Summary of average primary (PCEM), secondary (SCEM), and coincidence (COIN) rates accumulated in different test configurations in order to characterize dark count rates, UV suppression, and sensitivity to a very low-intensity beam

	Test Notes	Duration	PCEM [Hz]	SCEM [Hz]	COIN [Hz]
Dark Counts	ESA: -100 V	16 min	0.016 ± 0.0042	0.029 ± 0.0055	0.0036 ± 0.0019
	ESA: $-10,240$ V	15 min	0.023 ± 0.0051	0.065 ± 0.0085	0.0064 ± 0.0027
	ESA: HVSCI mode (Before UV test)	10 min	0.015 ± 0.0051	0.029 ± 0.0069	0.0018 ± 0.0019
	ESA: HVSCI mode (After UV test)	10 min	0.013 ± 0.0047	0.025 ± 0.0065	0.00
UV Test	Dwell at $+170^\circ$ az	1 min	0.18 ± 0.055	0.37 ± 0.079	0.00
	Dwell at $+170^\circ$ az	1 min	0.14 ± 0.048	0.18 ± 0.055	0.00
Weak Beam	ESA: HVSCI Mode (12 s sweeps)	30 min	0.47 ± 0.016	0.63 ± 0.019	0.17 ± 0.0096

**Fig. 17** Absolute detection efficiency as a function of ESA voltages (V) from continuous energy tests with H^+ and He^+ beams through the open aperture ($+90^\circ$ azimuth, 0° elevation). Dark curves highlight the required energy range (blue and green for SW and PUIs, respectively) while the lighter colors curves show the full measurable range. The dotted horizontal line is included for reference to indicate 11% detection efficiency

Energy Range Coverage The continuous energy tests, shown in Fig. 17, were performed with H^+ and He^+ beams entering the open aperture at $+90^\circ$ azimuth and 0° elevation. The required energy ranges for SW and PUIs are highlighted in dark blue and dark green,

respectively. The broader light blue and light green curves represent the full energy range measurable by SWAPI, demonstrating that the instrument exceeds its required coverage with substantial margin.

Detection Efficiency SWAPI's detector configuration also allows for an absolute detection efficiency calibration that is independent of variability in the incident flux. The efficiency is calculated using the relation:

$$\varepsilon = COIN^2 / (PCEM \times SCEM) \quad (1)$$

where PCEM, SCEM, and COIN are the primary, secondary, and coincidence count rates, respectively (Funsten et al. 2005; Elliott et al. 2016). As evident in Fig. 17, SWAPI's detection efficiency differs slightly between H^+ and He^+ , with average values of $\varepsilon_{H^+} = 0.1162 \pm 0.0129$ and $\varepsilon_{He^+} = 0.1221 \pm 0.0131$, respectively. While SWAPI does not perform mass discrimination via time-of-flight analysis, species-dependent variations in detection efficiency offer some qualitative means to distinguish between protons, alphas, and even heavier species (if present), as was the case for SWAP (Ebert et al. 2010). But more importantly, these efficiency differences are accounted for in the data analysis algorithms to ensure accurate interpretation of the various fluxes (see, e.g., Eq. (6) in Sect. 5.3).

Energy Resolution SWAPI's energy resolution was also characterized via continuous energy tests and independently confirmed using measurements from the full-width half-max fits to the energy-angle passbands (e.g., fits to counts vs. energy at fixed elevation). The resolution was found to range between 7% and 8.5%, with an average of $8.1 \pm 0.1\%$. The results exhibited a linear trend, with finer resolution observed at lower beam energies, and demonstrated that SWAPI meets its energy resolution requirements over the full operational range.

Analyzer Constant (k-Factor) The electrostatic analyzer constant, or *k-factor*, defines the proportional relationship between a particle's energy-per-charge and the ESA voltage required for its selection. Derived from the same data used to assess energy resolution, SWAPI's k-factor was found to be consistently between 1.90 and 1.95 across its full energy range for both H^+ and He^+ , validating the voltage-to-energy conversion used in flight operations (see Sect. 5.5, Table 9).

Sunglasses Attenuation Factor The attenuation factor of SWAPI's sunglasses aperture grid was empirically determined using data from the high-resolution azimuthal-angle response tests. Results from a 1 keV H^+ test are shown in Fig. 18. During this test, SWAPI was rotated from -170° to $+170^\circ$ at $0.1^\circ/s$, enabling continuous measurements of beam intensity across the full entrance aperture. Coincidence rates in the open aperture region ($\pm 40^\circ$ to $\pm 120^\circ$) were compared to those in the sunglasses region ($\pm 12^\circ$), yielding an attenuation factor of 1212 ± 25 – slightly exceeding the $\sim 1000\times$ design (Sect. 3.1.1) and demonstrating robust suppression performance. The count rates within the sunglasses region also remained flat – despite the low statistics – indicating uniform transmission and minimal scattering. These results validate the effectiveness of the electroformed nickel grid and vane configuration in suppressing flux in the SW-facing aperture region.

PUI Open Aperture Area The effective open aperture area for PUIs was determined by analyzing the response curve near the sunglasses and strongback regions, derived from the azimuthal-angle test data shown in Fig. 18. Effective aperture edges were identified through

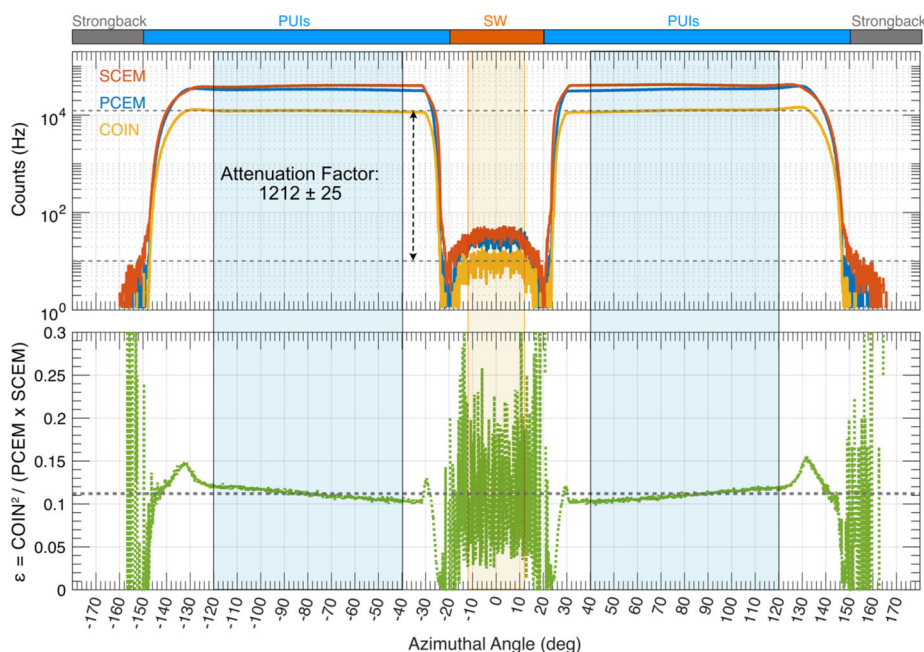


Fig. 18 Primary (PCEM), secondary (SCEM) and coincidence (COIN) rates accumulated by SWAPI during an azimuthal-angle roll from -170° to $+170^\circ$ at 0.1° s^{-1} in a 1 keV H^+ beam at 0° elevation, with the ESA voltage fixed to -504.9 V . The top panel shows the raw count rates while the bottom panel shows the computed absolute detection efficiency. Grey, blue, and orange bars at the top of the plot indicate the mechanical extent of the strongback, open aperture (PUIs), and sunglasses attenuation region (SW), with the sunglasses centered on 0° azimuth. Shaded regions highlight SW and PUI effective areas (orange and blue, respectively)

quantitative analysis of the vignettted regions near the vanes and strongback (Fig. 19), using a piecewise cubic Hermite interpolation to model the transition from full transmission to full obstruction. The effective angle was defined as the point where the integrated area under the response curve on either side of the roll-off was equal, yielding an equivalent open aperture – an angular range that preserves the same particle acceptance as a fully unobstructed aperture of equal area. This method produced an effective aperture of $220.85^\circ \pm 0.09^\circ$, in agreement with the expected value of 221° , confirming the accuracy of the instrument's true acceptance geometry.

Instrument Alignment Instrument alignment was verified through a combination of electro-optical testing and laser-tracked mechanical measurements. Azimuthal-angle response tests enabled precise mapping of effective aperture edges – based on vignetting roll-offs near the vanes and strongback as described above – while symmetry across all elevation angles was confirmed via energy-angle passbands at $\pm 90^\circ$ azimuth. Laser alignment checks performed prior to spacecraft integration demonstrated that SWAPI's internal alignment meets the required $\pm 0.25^\circ$ -per-axis tolerance between the instrument boresight and its mounting features. Measurements of key surfaces – including the top hat top plate, the mounting bolt circle, and vane edges – showed strong agreement with electro-optical results. For example, the effective aperture edge method found the sunglasses boresight offset by $\pm 0.14^\circ$, corresponding to a total of 0.28° between the vanes, closely matching the laser-tracked offset of

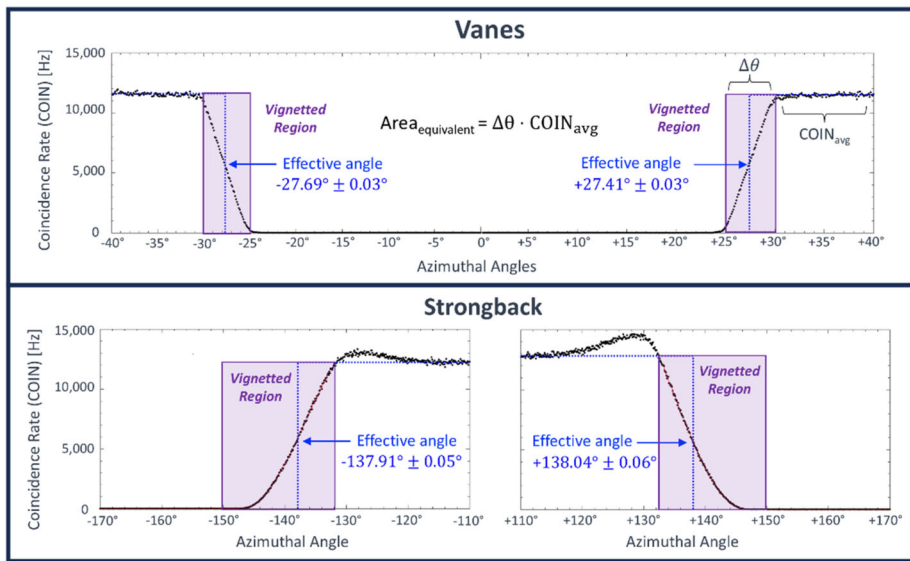


Fig. 19 Illustration of the effective aperture edges near the vanes (top) and strongback (bottom), used to characterize the extent of SWAPI's open aperture. The listed values in the plot represent the effective aperture edge angles

0.286° (0.206° and 0.0802° for the $+X$ and $-X$ vanes, respectively). These results confirm that SWAPI's boresight is well within specification, and that both alignment methods consistently validate the instrument's pointing knowledge – providing high confidence in the FOV orientation that feeds into the models and attesting to the high mechanical precision and workmanship achieved in the instrument's assembly.

Laboratory vs. Flight-Like Gain Curves Laboratory gain curves for the PCEM and SCEM, shown in Fig. 20, were obtained using 1 keV H^+ beams. These curves span the full CEM voltage range and were used to determine optimal operating voltages. The flight gain curves cover a narrower voltage range and are discussed further as a part of on-orbit calibrations in Sect. 4.3.

4.2 Cross-Calibration with CoDICE

Cross-calibration between SWAPI and CoDICE (Livi et al. 2025) took place from June 2 to June 10, 2025, following the completion of IMAP's environmental testing. As with the SWAPI calibration, all instrument operations were failure free. The primary objective was to compare measured energies and angles from both instruments within their overlapping energy range (approximately 0.5 to 20 keV), using a common He^+ beam. This effort aimed to establish a framework for mitigating instrument-specific discrepancies ahead of the more complex measurements in space (McComas et al. 2025b).

Testing was carried out at Princeton's Space Physics Laboratory, with both instruments mounted in the same vacuum chamber alternately taking measurements, as depicted in Fig. 21 (see also Fig. 43 in Appendix B). Table 7 summarizes the various combinations of energy and angle settings. Parameters were selected to ensure that both instruments could

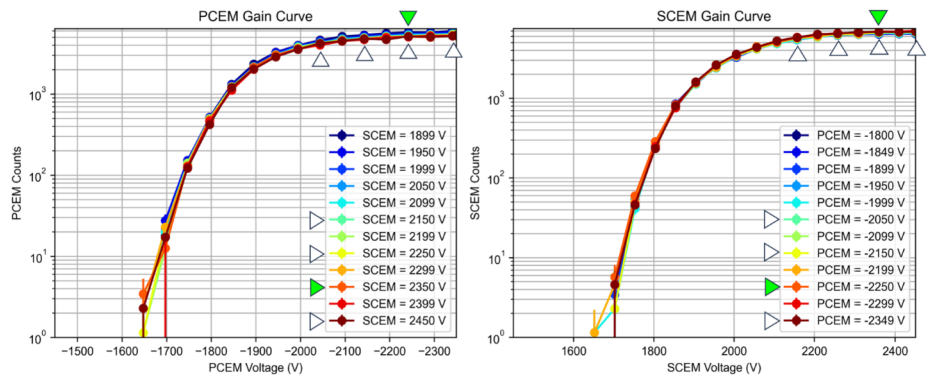


Fig. 20 Gain curves for PCEM (left) and SCEM (right) using 1 keV H^+ , with Poissonian uncertainties included (most are smaller than the points). The green arrows show target voltages selected for HV commissioning. White arrows indicate voltage steps used for the equivalent flight gain curve stepping scheme, described in Sect. 4.3

capture energy distribution measurements from the same direction for at least one energy and at least two elevation angles for each fixed azimuthal FOV.

Beam energies were fixed at 2 keV (for the 0° azimuth configuration) and 16 keV (for the 90° azimuth configuration). Instruments operated in both coarse and fine scanning modes, with voltage steps sufficient to sample the beam (see Sects. 6.2 & 6.4 for SWAPI; Livi et al. 2025 for CoDICE). The following ESA modes were used:

- Fine-step ESA voltage sweeps targeting 2 keV and 16 keV peaks:
 - CoDICE in reduced GF and nominal mode
 - SWAPI in HVENG mode (2 keV He^+)
 - SWAPI in HVSCI mode (16 keV He^+)
- Flight-like voltage sweeps using logarithmically spaced steps across the full energy range.

Elevation-angle mapping was achieved through a combination of coarse (1°) and fine (0.1°) scans, as listed in Table 7. Azimuthal orientations were fixed by the mounting hardware, so in order to target both the PUI (90° azimuth) and SW (0° azimuth) sectors, the chamber was vented mid-way through the test campaign, instruments were removed and rotated, and the entire assembly was re-installed.

4.3 On-Orbit Calibrations

SWAPI's on-orbit calibrations consist of high-voltage gain curves and in-flight comparison with the CoDICE instrument. SWAPI executes periodic gain tests using onboard Macro 5, which steps the PCEM and SCEM HVs from -200 V to $+100$ V around their nominal values. These tests are conducted while the ESA executes its standard 72-step sweep in HVSCI mode, producing 1-min measurements at 16 voltage combinations (Fig. 22). After each test, voltages return to their nominal settings. Based on the results, occasional adjustments to the target CEM voltages are made. During commissioning, initial CEM voltage settings are informed by lab-based gain curves (see Sect. 4.1, Fig. 20) for the science phase of the mission, on-orbit gain tests are performed every ~ 3 to 6 months to characterize and optimize performance as the detectors age. One challenge in interpreting the gain test results is due to the variability of SW count rates. To mitigate this, the voltage profile is repeated four times over an hour, allowing exclusion of outlier measurements for any given voltage combination.

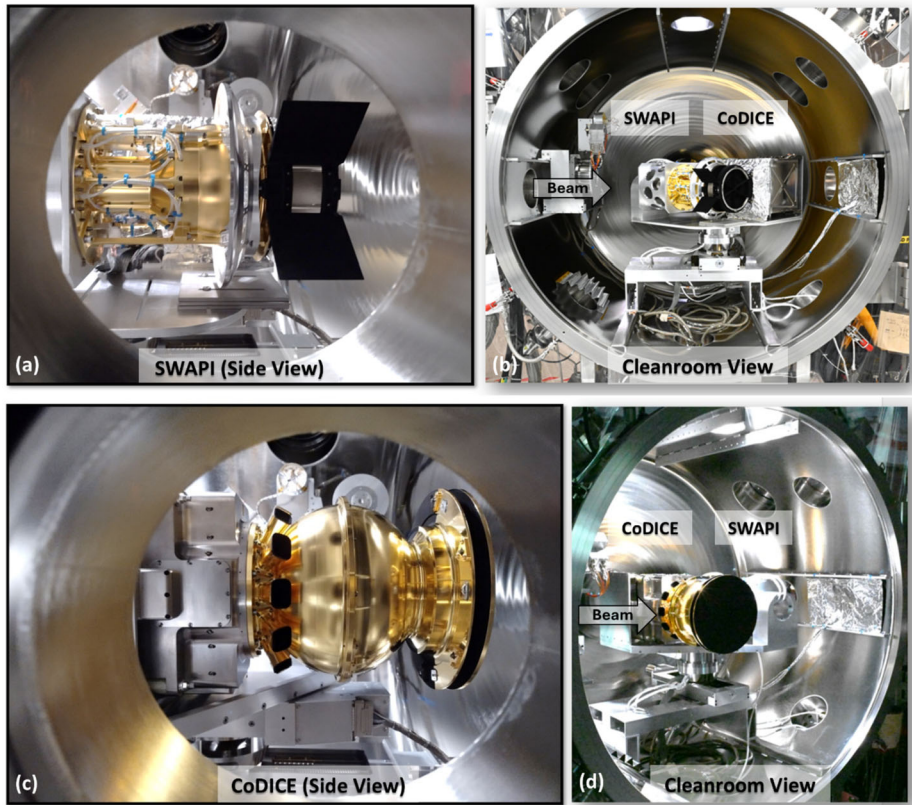


Fig. 21 Photographs showing the SWAPI and CoDICE instruments mounted together in the Princeton Main Chamber to alternately share the beam. Panels (a) and (c) show the instruments under vacuum, oriented toward a viewport flange on the chamber's side, away from the beam. Panels (b) and (d) show the chamber is vented to atmosphere from the cleanroom side with the door open. During this phase of testing, the instruments were configured to calibrate their SW sectors using a 2 keV He⁺ beam

In-flight comparisons with CoDICE are performed routinely using validated data from both instruments. These comparisons require no special commanding; both instruments operate in their nominal science modes, collecting simultaneous observations. Ground-based processing is used to analyze the results across their overlapping energy range.

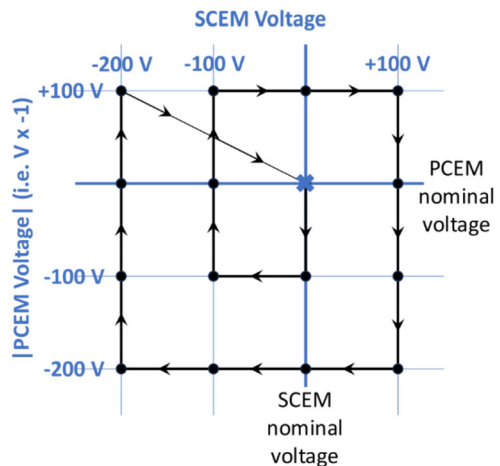
5 Sensor Performance Model

A comprehensive sensor performance model was developed during SWAPI's design phase to support instrument optimization, algorithm development, calibration planning, and validation of look up tables (LUTs) used in data processing. The model simulates the full end-to-end response of the instrument using real, historical SW observations as inputs and compares the processed outputs to the known inputs, enabling a closed-loop validation of the model's performance.

Table 7 Summary of parameters used during SWAPI-CoDICE cross-calibration. Target beam energies included 16 keV He⁺ through SWAPI's open aperture (90° azimuth; PUI configuration) and 2 keV He⁺ through the SWAPI sunglasses (0° azimuth; SW configuration). For each elevation angle setting, translational adjustments were made to maintain a consistent beam radius at the aperture for each instrument

Species	Beam Energy	ESA Target Energies	Instrument Operational Modes	Elevation Angle		Azimuthal Angle
				SWAPI	CoDICE	
Helium (He ⁺)	2 keV 16 keV	2 keV (peak) <i>sunglasses</i> 16 keV (peak) <i>open aperture</i>	CoDICE: reduced GF & nominal modes SWAPI: HVENG CoDICE: reduced GF & nominal modes SWAPI: HVSCI	-5° -2° 0° +2° +4° +7° +8°	0° +2° +4° +7° +8°	0° 90°
				<u>fine scans:</u> -8° to -5° in 0.2° steps; -5° to +5° in 0.4° steps; and +5° to +8° in 0.2° steps	<u>fine scans:</u> -4°, -3°, -2°, -1° to +5° in 0.2° steps, and +6° to +9° in 0.2° steps	
		Flight-Like Sweeps	Nominal flight sweeping schemes		0°	

Fig. 22 SWAPI Gain Test



5.1 SWAPI's End-to-End Model

SWAPI's end-to-end model comprises the following interconnected modules: (1) inputs for SW OMNI observations (e.g., ACE and Wind at 1 au), (2) a SW and PUI distribution function model, (3) coordinate system transformation matrices, (4) a Geant4-based entrance aperture simulation, (5) a SIMION[®]-based electro-optical model, and (6) a telemetry-formatting module to convert simulated outputs to expected instrument data products. A diagram of these modules is shown in Fig. 23. The model follows a “V-diagram” structure: the left branch simulates the physical environment and instrument response, while the right-hand branch emulates ground-based data processing. In-flight data processing mirrors the

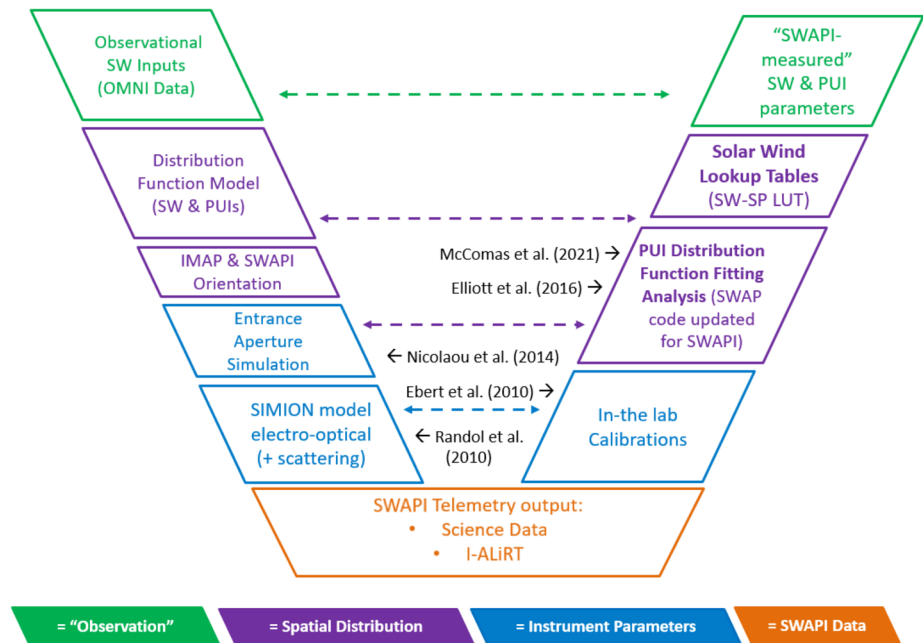


Fig. 23 Summary of the SWAPI end-to-end model

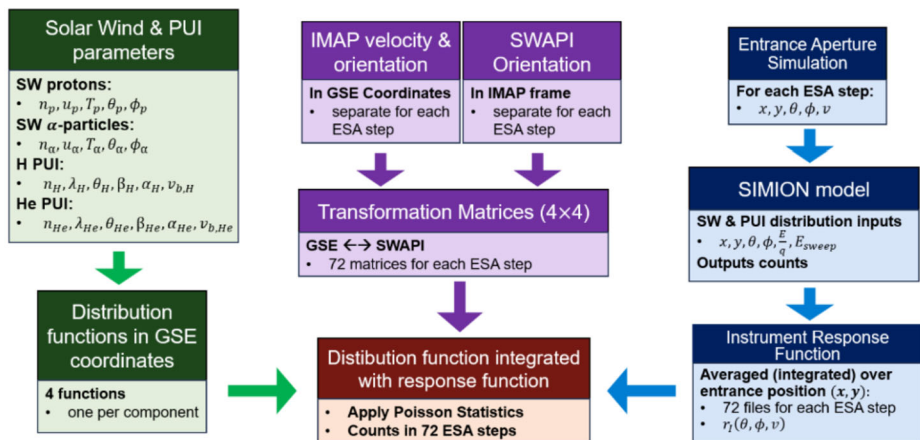


Fig. 24 Flow diagram and summary of parameters generated by various modules of the SWAPI end-to-end model

right branch, with minor differences: raw telemetry (orange) corresponds to Level 0 data, laboratory calibrations (blue) are supplemented by in-flight calibrations, and distribution function fitting is performed algorithmically. Figure 24 summarizes the parameters generated by each module.

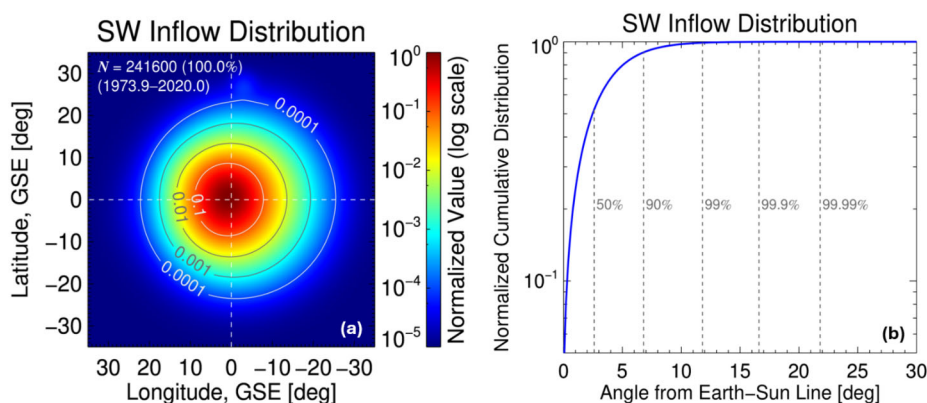


Fig. 25 Normalized SW inflow distribution as a function of longitude and latitude in GSE coordinates for: (a) observed T_p and (b) the resulting cumulative distribution

5.2 Solar Wind OMNI Observations

To inform the design of SWAPI's entrance aperture and angular acceptance, a data-driven analysis was performed using long-term SW observations from the OMNI database (King and Papitashvili 2005). The analysis examined the angular distribution of SW inflow directions in Geocentric Solar Ecliptic (GSE) coordinates. Two modeling assumptions were used: (i) a finite-temperature model that assumed a 2D Gaussian angular distribution based on measured SW temperature, and (ii) a cold beam limit that assumed negligible thermal spread ($T_p \rightarrow 0$) approximated by a fixed angular width ($\sigma_p = 0.1^\circ$).

The analysis used $\sim 241,600$ hourly OMNI data points that met the following criteria:

- Availability of SW proton density, temperature, and flow speed
- Availability of flow angles (latitude and longitude)
- Presence of the alpha-to-proton ratio (used only as a selection filter)

OMNI flow angles are defined in a modified GSE coordinate system: latitude indicates the direction the SW is traveling, and longitude is negative for from the direction it originates. These are inverted to represent inflow directions in standard GSE coordinates. Aberration corrections for spacecraft motion are also applied.

For each OMNI data point, the SW angular distribution is modeled using:

$$f_p(\theta, \varphi) \propto \frac{n_p}{\sigma_p^2} \times \exp\left(\frac{-(\theta - \theta_b)^2 - (\varphi - \varphi_b)^2}{2\sigma_p^2}\right) \quad (2)$$

where: $\sigma_p = \frac{v_{th,p}}{u_p} \times \frac{180^\circ}{\pi}$ and $v_{th,p} = \sqrt{\frac{2k_B T_p}{m_p}}$. This function is evaluated on a 2D angular grid (θ, φ) for each time step, using the measured bulk flow direction (θ_b, φ_b) . In the cold-beam limit ($T_p \rightarrow 0$), the same analysis is repeated with a fixed angular width ($\sigma_p = 0.1^\circ$) to isolate the core inflow direction. Figure 25a shows the angular distribution for the finite-temperature model, summed over all OMNI data times and normalized to the peak value, while Fig. 25b quantifies the fraction of particles arriving within a given angle of the Earth-Sun line. These results were critical in defining the angular width of the sunglasses.

5.3 Solar Wind & PUI Distribution Function Module

The SW H^+ distribution is modeled using a Maxwellian distribution:

$$f(v) = n_p \left(\frac{1}{\pi v_{th}^2} \right)^{3/2} \exp \left(-\frac{v^2}{v_{th}^2} \right), \quad (3)$$

where v is the particle speed in the plasma frame, n_p is SW proton density and $v_{th} = \sqrt{\frac{2k_B T_p}{m_p}}$ is the thermal speed, which includes Boltzmann's constant, k_B , SW proton temperature, T_p , and proton mass, m_p . The particle speed in the plasma frame is computed as:

$$v = |(\mathbf{R}\mathbf{v}' + \mathbf{V}_{IMAP}^{GSE}) - \mathbf{u}_p^{GSE}|, \quad (4)$$

for particle velocity $\mathbf{v}'$ in the SWAPI frame, IMAP velocity $\mathbf{V}_{IMAP}^{GSE}$, and SW velocity $\mathbf{u}_p^{GSE}$ in the GSE frame, and rotation matrix $\mathbf{R}$ from SWAPI to the GSE frame (see Sect. 5.4 for more details on the transformation from the SWAPI frame). In the GSE frame, the SW velocity is:

$$\mathbf{u}_p^{GSE} = \begin{pmatrix} -u_p \cos(\phi_p - \Phi_{OS}) \cos(\theta_p) \\ u_p \sin(\phi_p - \Phi_{OS}) \cos(\theta_p) \\ u_p \sin(\theta_p) \end{pmatrix} \mp \mathbf{V}_{IMAP}^{GSE}, \quad (5)$$

where Φ_{OS} is IMAP's Off-Sun pointing angle, u_p is the SW flow speed, (θ_p, ϕ_p) are the SW flow angles relative to the SWAPI radial vector (longitude, latitude), and $\mathbf{V}_{IMAP}^{GSE}$ is IMAP's velocity, in GSE. The forward model uses input parameters $(u_p, \theta_p, \phi_p, n_p, T_p)$ to simulate SWAPI count rates over each 12 s sweep. The Maxwellian distribution is converted to count rates using the full instrument response function:

$$C(E_e) = \frac{\varepsilon \frac{G(E_e)}{2} \times \int_{E_{min}}^{E_{max}} \int_{\theta_{min}}^{\theta_{max}} \int_{\phi_{min}}^{\phi_{max}} R(E, \theta, \phi) f(V, \theta, \phi) V^4 dE \cos \theta d\theta d\phi}{\int_{E_{min}}^{E_{max}} \int_{\theta_{min}}^{\theta_{max}} \int_{\phi_{min}}^{\phi_{max}} dE \cos \theta d\theta d\phi}, \quad (6)$$

where E_e is ESA energy, $G(E_e)$ is the energy-dependent geometric factor, θ is the elevation angle, ϕ is the azimuthal angle, and $R(E, \theta, \phi)$ is the instrument response function (derived from SIMION®, Geant4, and lab calibrations), and ε is the detector efficiency, and V is the particle speed in the SWAPI frame.

The SW $^4He^{++}$ distribution is also modeled using a Maxwellian with input parameters $(u_\alpha, \theta_\alpha, \phi_\alpha, n_\alpha, T_\alpha)$. For simplicity, the alphas are assumed to be co-moving with the protons, such that $u_\alpha = u_p$, $\theta_\alpha = \theta_p$, $\phi_\alpha = \phi_p$. However, this assumption does not always hold at 1 au and is relaxed in the Level 3 data processing (see Sect. 7.3.2). Even in such cases, only the alpha flow speed is derived (not the deflection angles), due to the statistical limitations of the data.

Table 8 Summary of parameters used in the generalized filled-shell distribution model ($k = H$ or He)

Symbol	Description	Value/Notes
$\alpha_{PUI, k}$	Cooling index (=1.5 for adiabatic cooling)	
w_k	PUI speed÷local cutoff speed (SW frame)	$w_k = v/v_{b, k}$
$v_{b, k}$	Local cutoff speed: difference of SW bulk velocity ($\mathbf{u}_{SW}$) and interstellar neutral bulk velocity ($\mathbf{u}_k$) (solar inertial frame)	$v_{b, k} = \mathbf{u}_{SW} - \mathbf{u}_k $
$\beta_{E, k}$	ionization rate at $r_E = 1$ au (solar cycle average)	$5 \times 10^{-7} \text{ s (H}^+)$ $1 \times 10^{-7} \text{ s}^{-1} \text{ (He}^+)$
$n_k(r w_k^{\alpha_{PUI, k}}, \psi)$	Density of interstellar neutrals, obtained by the hot model	Thomas (1978)
$n_{TS, k}$	Densities at the termination shock (for hot model)	$n_H = 0.127 \text{ cm}^{-3}$ $n_{He} = 0.015 \text{ cm}^{-3}$
$T_{TS, k}$	Temperature at the termination shock (for hot model)	$T_H = 10,000 \text{ K}$ $T_{He} = 7500 \text{ K}$
ψ	angle between radial and interstellar inflow	
λ	ionization cavity size for neutral H or He	$\lambda_H = 4.0 \text{ au}$ $\lambda_{He} = 0.5 \text{ au}$
$\theta_{ISN}, \phi_{ISN}, u_k$	Interstellar inflow (longitude, latitude, speed) in Ecliptic J2000 coordinates; hydrogen from Lallement et al. (2005); helium from McComas et al. (2015)	Hydrogen (252.2° , 9.0° , 22 km s^{-1}) Helium (255.7° , 5.1° , 25.4 km s^{-1})
Θ	Heaviside step function	

PUI populations (H^+ and He^+) are modeled using the generalized filled-shell distribution (Chen et al. 2014; Vasyliunas and Siscoe 1976):¹

$$f(r, w_k, \psi) = \frac{\alpha_{PUI, k}}{4\pi} \frac{\beta_{E, k} r_E^2}{r u_{sw} v_{b, k}^3} w_k^{\alpha_{PUI, k}-3} n_k(r w_k^{\alpha_{PUI, k}}, \psi) \Theta(1 - w_k), \quad (7)$$

for PUIs at position (r, ψ) in the solar inertial frame with parameters summarized in Table 8. As with the SW described above, SWAPI's PUI counts are modeled using 4 sets of input parameters applied to Eq. (6): $(n_k(r, \psi), \beta_{E, k}, \alpha_{PUI, k}, v_{b, k})$.

5.4 Coordinate System Transformation Matrices

To integrate the OMNI SW measurements into SWAPI's end-to-end model framework, a set of coordinate system transformation matrices was developed to convert between SWAPI's reference frame and the GSE frame used by OMNI (specifically, a modified, or “quasi” GSE system; see below). The instrument coordinate system is defined so that longitude = 0° and latitude = 0° corresponds to the center of SWAPI's FOV. In this reference frame, the spacecraft spins about the instrument's x-axis and the y-axis rotates towards the z-axis in the direction of positive spin (see Sect. 4.1, Fig. 16 for instrument coordinates).

¹In principle, the gradient in neutrals for H^+ is large at 1 au, rendering the filled shell approach inaccurate at L1 (e.g., Gloeckler and Geiss 1998; Schwadron 1998, and Schwadron et al. 2000). However, the approach is still applicable for He^+ (Sect. 7.3.3) which is the primary focus of SWAPI's PUI data products. Nevertheless, the treatment of H^+ will likely be re-visited for rare conditions when it can be distinguished from the SW $^4\text{He}^{++}$ peak.

Table 9 Comparison of various performance parameters derived from calibration measurements vs. the SWAPI SIMION model. The flight calibration values listed for energy resolution and k-factor reflect the average across SWAPI's full energy range

Parameter	Flight Calibration	SWAPI SIMION Model
Energy Resolution (dE/E)	$8.1 \pm 0.1\%$	8.5%
k factor [V/eV]	1.93	1.88
Aperture Area [cm ²]	1.742 ± 0.053 cm ²	1.703 ± 0.003 cm ²

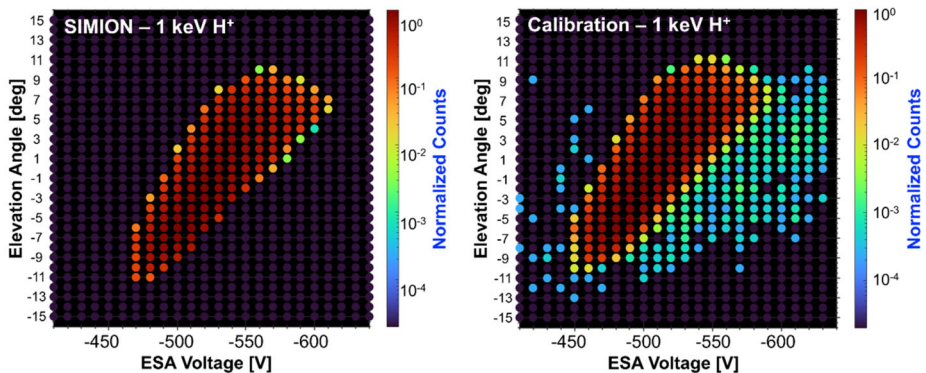


Fig. 26 Energy-angle passband comparison of (left) modeled and (right) measured cases of a 1 keV H⁺ beam entering through the open aperture region, with parameters matched to the SWAPI flight calibration measurements. Data taken from the real flight instrument also show low levels of actual particle scattering in the instrument (blue and green points); which are not included in the model

5.5 Electro-Optical Model

SWAPI's electro-optical model builds on the one developed for SWAP (McComas et al. 2008; Randol et al. 2010; Nicolaou et al. 2014) with a several minor changes to the geometry file, including: (a) removal of the RPA and deflector ring electrodes, (b) addition of grounded electrodes to reflect the aperture cutoff ring, (c) an update of the Inner ESA Dome geometry that incorporates the 0.040" rounded edge (recall Sect. 3.4.1), and (d) revised ESA voltages consistent with SWAPI's flight configuration. Electrostatic modeling was performed using SIMION[®] 8.1, which solves the Laplace equation for user-defined 2D or 3D electrode geometries using finite-difference methods (Dahl 2000). Aside from these separately modeled passive elements, the electro-optical model geometry closely matches the as-built flight design (recall Fig. 5). Table 9 compares the energy-per charge resolution, k factor, and aperture area achieved through SIMION simulations vs. measurements taken during the SWAPI flight instrument's calibration campaign (Sect. 4.1.2), while Fig. 26 illustrates the comparison of energy-angle responses achieved via SIMION[®] to those taken during calibration in the lab (Sect. 4.1.1). The example shown here is for 1 keV H⁺ (measured through the open aperture rather than the sunglasses, for statistical reasons). These comparisons exemplify the excellent agreement between the flight instrument's calibration data and SIMION[®] model.

Figure 27 presents the modeled results for SWAPI's aperture area, a key parameter used in the calculation of the instrument's effective geometric factor (recall Table 2 of Sect. 2). The detection region was mapped by using a fixed-energy, uniform parallel beam (1 keV

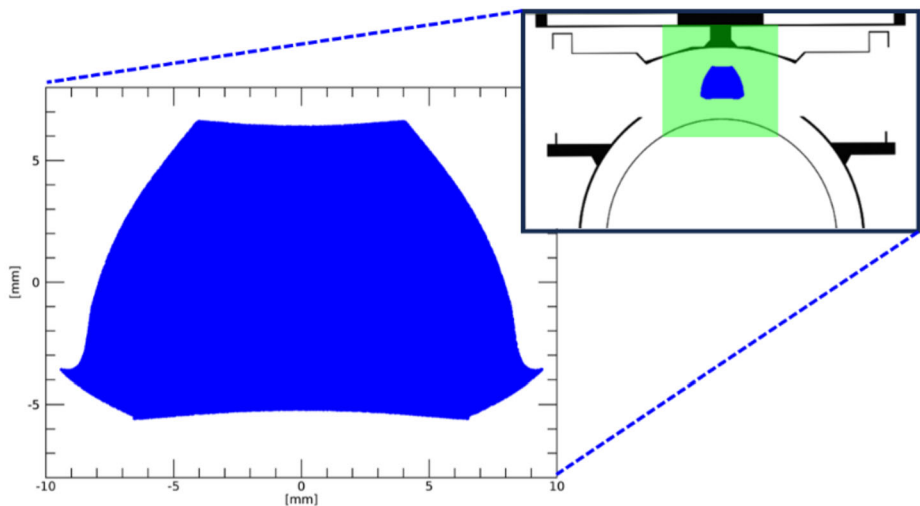


Fig. 27 Simulation results for SWAPI's aperture area, showing the region of particles that get detected at the foil (blue) from an initial beam of 1 million particles (in this case, 1 keV H^+) uniformly filling the instrument's aperture (green region). Analysis yielded an aperture area of $1.703 \pm 0.003 \text{ cm}^2$

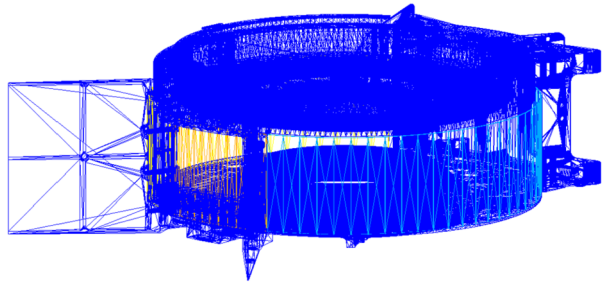
H^+) of one million particles that filled the instrument's aperture. Its area was then quantified by setting the ESA voltage to the appropriate level, tracking the number of particles that reached the foil, and tracing them back to their points of incidence on the aperture. The result, converted to real-world coordinates, was: $A_{\text{SWAPI}}(\text{simulation}) = 1.703 \pm 0.003 \text{ cm}^2$. A laboratory-based measurement of this parameter was independently achieved during the SWAPI-CoDICE cross-calibration, by using the absolute beam monitor (ABM) as a reference to measure absolute flux of the beam. The measurement-derived value was: $A_{\text{SWAPI}}(\text{Calibration}) = 1.742 \pm 0.053 \text{ cm}^2$, showing strong agreement with the simulations. Given that both values are statistically consistent within their respective uncertainties, the simulated result is adopted for algorithm development due to its higher statistics and easier reproducibility.

Lastly, SIMION[®] was used to characterize SWAPI's response to ions across the full range of incident entry locations, angles, energies and ESA voltages. For each run, the simulation was configured with flight-like values for the ESA and CEM voltages (recall Fig. 5). Ions were launched with varying initial conditions to span the instrument's full acceptance range, and their trajectories were tracked to determine which particles reached the foil – the critical location for initiating coincidence detection in flight. From these simulations, a comprehensive SWAPI response function was constructed, capturing the instrument's sensitivity to ion energy and angle at each ESA voltage. This function was then integrated under the assumption of a uniform incident beam to generate one of the core LUTs used to develop and validate SWAPI's data-processing algorithms (Sect. 7.3).

5.6 Entrance Aperture Simulation

Geant4, a Monte-Carlo based toolkit that simulates the passage of particles through matter (Agostinelli et al. 2003; Allison et al. 2006, 2016), was employed to simulate the more complex geometry of SWAPI's entrance aperture (Fig. 28). This approach enables accurate modeling of SWAPI's 3D features – including the sunglasses, vanes, and strongback

Fig. 28 Geant4 Model of SWAPI aperture, visualized using the HepRApp software. The cyan-shaded aperture represents the open aperture, PUI region, while the orange-shaded aperture represents the “sunglasses” SW region. Other instrument structures through which particles cannot enter are outlined by the wireframes in blue



– without breaking the cylindrical symmetry employed to achieve computational efficiency in SIMION®. In the simulation, particles were launched from a hemispherical surface surrounding the instrument’s aperture, with a cosine-distributed profile. Energy, position and direction were recorded for each particle, immediately after passing through the entrance. These outputs define the 3D distribution of particles used to establish the initial conditions in SIMION®, while enabling realistic modeling of SWAPI’s FOVs and geometric response. The Geant4 module also includes capabilities to simulate scattering, energy loss, and secondary particle production, which may be leveraged in future studies to investigate background contributions from sources such as cosmic rays.

5.7 Application: SWAPI’s Aperture Design Trade

The final design of SWAPI’s aperture grid assembly – from the electroformed nickel grid (Sect. 3.1.1) to the pair of vanes – was informed by a combination of experimental constraints (e.g., CEM count rate limits, instrument geometry) and forward modeling of the SW and PUI environments at 1 au. The $\sim 0.1\%$ attenuation of the SW flux was selected to match the dynamic range of SWAP. In addition, a comprehensive trade study was conducted to optimize the geometry and orientation of the aperture components, with dual goals of enabling high-fidelity SW measurements while also maximizing acceptance of PUIs, particularly at high energies near the He^+ cut-off. The study utilized 65 years of OMNI data and determined that a 15° half-angle for the sunglasses, combined with precisely angled vanes, would keep the core SW protons within the aperture 99.99% of the time. The vane geometry was further tuned to suppress scattering, keeping the SW from entering the ESA unattenuated 99% of the time. This configuration ensures that the SW is sufficiently reduced to avoid overwhelming the much weaker PUI signal, while also retaining a large $\sim 221^\circ$ open-aperture region for unrestricted PUI entry (Sect. 4.1.2, Figs. 18 & 19).

5.8 Application: SWAPI’s Anticipated Performance at 1 au

The integrated end-to-end model was used before launch to simulate SWAPI’s expected performance at 1 au, as illustrated in Fig. 29. During nominal operations, SWAPI measures combined distributions of SW H^+ and $^4\text{He}^{++}$ at a 1-min cadence (5×12 s sweeps) and He^+ PUIs at a 10 min cadence (50×12 s sweeps). Like SWAP, SWAPI is also capable of detecting suprathermal ions and PUIs generated downstream of shocks and other transient structures (Zirnstein et al. 2018; McComas et al. 2022; Shrestha et al. 2024). In addition to informing the instrument’s early design, the end-to-end model also served as a critical tool for validating SWAPI’s data-processing algorithms (Sect. 7).

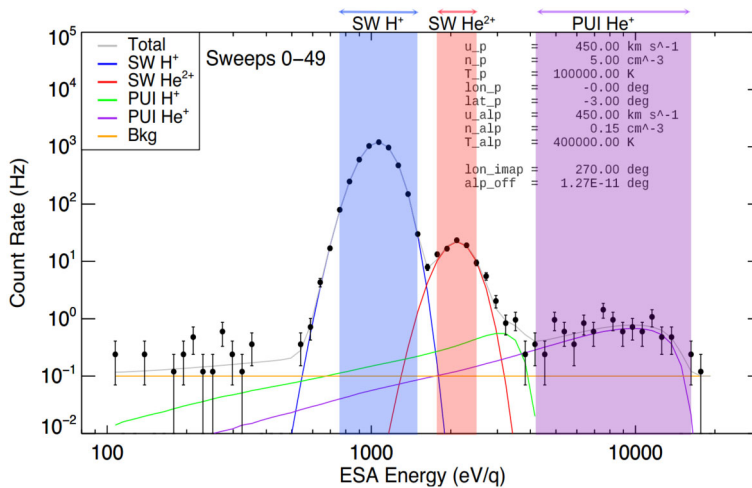


Fig. 29 Modeled SWAPI count rate distribution obtained by statistically combining 50 coarse sweeps (10 min cadence), with fit regions highlighted for SW protons (H^+ ; blue), alphas ($^4He^{++}$; red), and PUIs (purple). In flight, the 10 min time cadence data like those shown here are used to derive SWAPI's PUI parameters, while the SW parameters are returned at a 1 min cadence (e.g., the 5-sweep equivalent to this plot)

6 Instrument Operations

This section provides details about SWAPI's software states, operational modes, autonomy, science data acquisition, routine operations, and telemetry handling.

6.1 Software States

SWAPI operates in four software states:

- OFF – no power applied
- INITIAL – executed bootloader code from PROM upon power-up
- BOOT – transitional state for system checks and configuration (PROM)
- NORMAL (EEPROM) – Executes science and engineering operations from EPROM code loaded into RAM

Upon power-up, the instrument runs the PROM code for ~ 20 s before transitioning to the NORMAL state, where the primary science and engineering software executes.

6.2 Science and Engineering Modes

The NORMAL state includes four operational modes, listed in Table 10. Upon entry to the NORMAL state, low-voltage engineering (LVENG) is the default mode and Macro 0 runs automatically. The mode transition diagram is shown in Fig. 30.

6.3 Flight Software (FSW) Autonomy

SWAPI FSW autonomously transitions to SAFE mode under any of the following conditions:

Table 10 SWAPI's Science and Engineering Modes

Mode	Description	Commanding	Telemetry
LVENG	Low-voltage engineering. Default mode upon entering BOOT or NORMAL. Supports low voltage engineering and maintenance operations.	Memory load/dump, EEPROM write	Housekeeping, memory dump, event messages, autonomy
LVSCI	Low-voltage science mode. Enables full science acquisition functions without enabling high voltage. ESA pseudo-sweeps performed without setting the HV DAC. Stim pulsers can be enabled for end-to-end checks.	Science acquisition, stim pulser	Housekeeping, event messages, autonomy, real-time science
HVENG	High-voltage engineering: used for calibration, commissioning and HV turn-on. HV supplies are under direct command control; ESA sweeps are not automatically performed by the SWAPI FSW. PCEM, SCEM and COIN data collected and telemetered in the real-time science packet. Safing transitions immediately to LVENG.	HV commands, science acquisition, safing	Housekeeping, event messages, autonomy, real-time science
HVSCI	High Voltage Science: primary science acquisition mode. The ESA HV sweeps automatically with sweep levels defined by selectable LUTs. Safing transitions immediately to LVENG.	Science acquisition, safing	Housekeeping, event messages, autonomy, real-time science, I-ALIRT

- Two consecutive out-of-limits samples for:
 - Overtemperature (Control Board, HVPS, or Sensor)
 - Over/undervoltage (LVPS, PCEM, or SCEM)
 - Overcurrent (LVPS, PCEM, or SCEM)
- Six out-of-limits samples indicate exceeding count rate thresholds (1,344,882 Hz for PCEM and SCEM)
- Rejection of any macro command

Upon entering SAFE mode, the instrument switches to LVENG and asserts the Power Off Request flag, prompting the spacecraft to power down SWAPI.

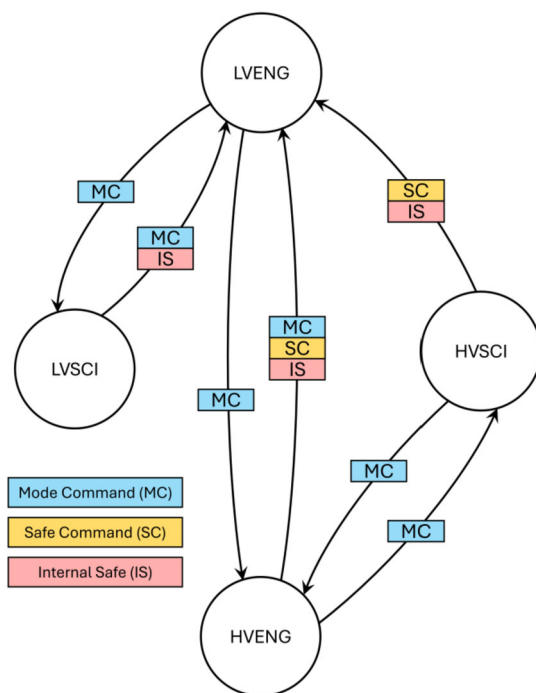
6.4 Science Data Collection

Science data are acquired in HVSCI mode, during which the ESA HVPS steps the voltage on the Inner ESA Dome according to a predefined profile selected from one of 64 stored plans in the instrument's Look-up Table (LUT). Each plan contains 32 steps, and each step can execute any of 16 sweep tables. Steps can be repeated up to 4095 times, allowing flexible control over data acquisition.

6.4.1 Sweep Table Configuration

The nominal sweep table consists of 72 steps:

- 1 large voltage transition step (not used for science)
- 62 coarse steps spanning 0.1 to 20 keV/q
- 9 fine steps centered on the peak E/q bin

Fig. 30 SWAPI's mode transitions

Each step lasts 1/6 second (≈ 166.667 ms), yielding a 12 s sweep cycle. Time allocation per step is: 16.667 ms for ESA voltage settling, 145 ms for data acquisition (“live time”), and 5 ms timing margin.

6.4.2 ESA Voltage Table and Indexing

Each sweep step corresponds to a micro-step in a 1024-entry ESA voltage table with log-linear spacing. Coarse steps use absolute indexing with 16 micro-steps between each. Fine steps use relative indexing, dynamically adjusted by the onboard peak-finding algorithm to center the fine sweep around the detected peak.

6.4.3 Peak-Finding Algorithm

During the coarse sweep, the FSW identifies the coarse step with the highest coincidence count rate (step n). The fine sweep then spans from $n - 16$ to $n + 16$ micro-steps, advancing in 4 micro-step increments, centering the fine sweep on the coarse peak.

Figure 31 shows the peak-finding algorithm in operation during a single ESA sweep, based on SWAPI's flight calibration data.

6.4.4 Flight Plans

Only two plans are intended for flight operations:

- Plan 1: Repeatedly executes Sweep Table 1, limiting ESA HV output to approximately -3 keV for HV commissioning.
- Plan 2: Repeatedly executes Sweep Table 2, utilizing the full ESA voltage range (-52 V to -10.24 keV) for routine science operations.

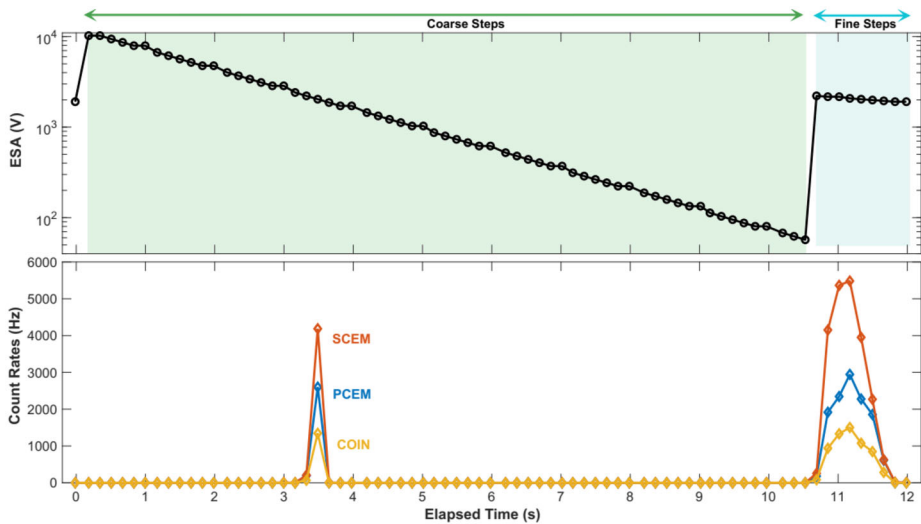


Fig. 31 Single 12 s full-range ESA sweep (top panel) and count rates for a 4 keV He^+ beam (bottom panel) through SWAPI's open aperture (90° azimuth). Individual data points reflect the combination of 62 coarse steps (shaded green) and 9 fine steps (shaded blue). The initial step is used for the voltage transition and is always discarded in the data analysis. In the bottom panel, the first peak shows SWAPI's initial detection of the beam, while the second peak demonstrates accurate detection and placement of fine steps, thereby achieving a higher-resolution beam profile. Data shown here was taken on 9/27/2024, during SWAPI's flight calibration campaign. For the flight data, five such sweeps are combined to achieve the required 1-min cadence of the SW distribution, and the fine steps enable even higher-resolution measurements of the solar wind proton peak

6.4.5 Science Data Compression

To reduce bandwidth, the Application Flight Software (AppFSW) implements a lossy compression scheme for science data. Although the hardware counters for PCЕМ, SCЕМ, and Coincidence rates are 24 bits, only 17 bits are allocated in telemetry: a 16-bit count and a 1-bit compression flag. Our scheme preserves the dynamic range while reducing the data volume, with overflow conditions explicitly flagged for ground-based reconstruction.

6.5 Routine Operations

SWAPI's nominal science operations are straightforward, consisting of continuous 72-step ESA voltage sweeps with fixed PCЕМ and SCЕМ voltage settings. However, routine operations are periodically interrupted during delta-V and station keeping maneuvers and a few times a year for the gain tests.

6.5.1 Gain Tests

As described in Sect. 4.3, SWAPI performs gain tests approximately every 3-6 months – more frequently during early mission phases via Macro 5 (see Fig. 22). Adjustments to the CЕМ bias voltages may be implemented to maintain optimal detector performance.

6.5.2 Operations During Maneuvers

To mitigate the risk of contamination from thruster plumes, SWAPI is powered off for one hour before, during, and for one hour after delta-V and station-keeping maneuvers. However,

Table 11 SWAPI Peak and Average Data Rates. *Non-science data. **Average rate at which data is written to the SSR in nominal ops. The real-time downlink does not include science packets, which reduces SWAPI's real-time data rate to 11.2 bps (housekeeping only)

Name	Bytes per Packet	Average Downlink (bps)	Average Daily on S/C Bus (bps)	Nominal Max Transfer on S/C Bus (bps)	Off-Nominal Max Transfer on S/C Bus (bps)
*Autonomy	12	n/a	96	96	96
*Housekeeping	84	11.2	11.2	672	672
*Memory Dump	150	n/a	n/a	n/a	1200
*Messages	18	n/a	n/a	n/a	144
Science	54	432	432	432	432
I-ALiRT	32	n/a	256	256	256
Total	n/a	443.2**	1456	1456	2800

the instrument remains on during the routine daily $\sim 1^\circ$ repointing maneuvers (McComas et al. 2025b; Hegarty et al. 2025), which pose negligible risk.

6.6 Telemetry Rates

SWAPI uses Telemetry Output (TO) tables to define which data packets are included in the real time downlink and at what cadence. In parallel, Packet Filter Tables (PFTs) determine which packets are stored on the spacecraft's Solid-State Recorder (SSR) for later downlink. Table 11 summarizes the expected peak and average data rates for both real-time and recorded telemetry under nominal and off-nominal operating conditions.

7 Data Products & Algorithms

SWAPI's data structure follows the standardized data product definitions established by the IMAP Science Operations Center (SOC) (Reno et al. 2025) for Level 0 to Level 3:

- Level 0: Reconstructed, unprocessed instrument, payload, and spacecraft data with all communications artifacts removed. These data are formatted as binary CCSDS packets.
- Level 1: Fully decommutated, uncalibrated raw data at full resolution, time-referenced and organized as extracted telemetry items.
- Level 2: Level 1 data converted to physical units using calibration, ancillary, and supporting data. These represent the lowest tier of research-grade scientific data and retain the same temporal and spatial resolution as Level 1.
- Level 3: Resampled data products, potentially merged with measurements from other instruments to produce derived quantities (e.g., density of a distribution).

SWAPI utilizes seven distinct types of data packets, (see Sect. 3.6, Table 4); reporting simultaneous observations of SW and PUIs via the science and I-ALiRT telemetry (Lee et al. 2025).

The raw science data consists of PCEM, SCEM, and Coincidence counts packaged in CCSDS telemetry packets. These packets have variable lengths, however, all SWAPI SCI packets are fixed at 54 bytes (432 bits), each representing 1 s of data (6 ESA steps). All the packet types include a CHKSUM parameter calculated both onboard and on the ground to verify data integrity.

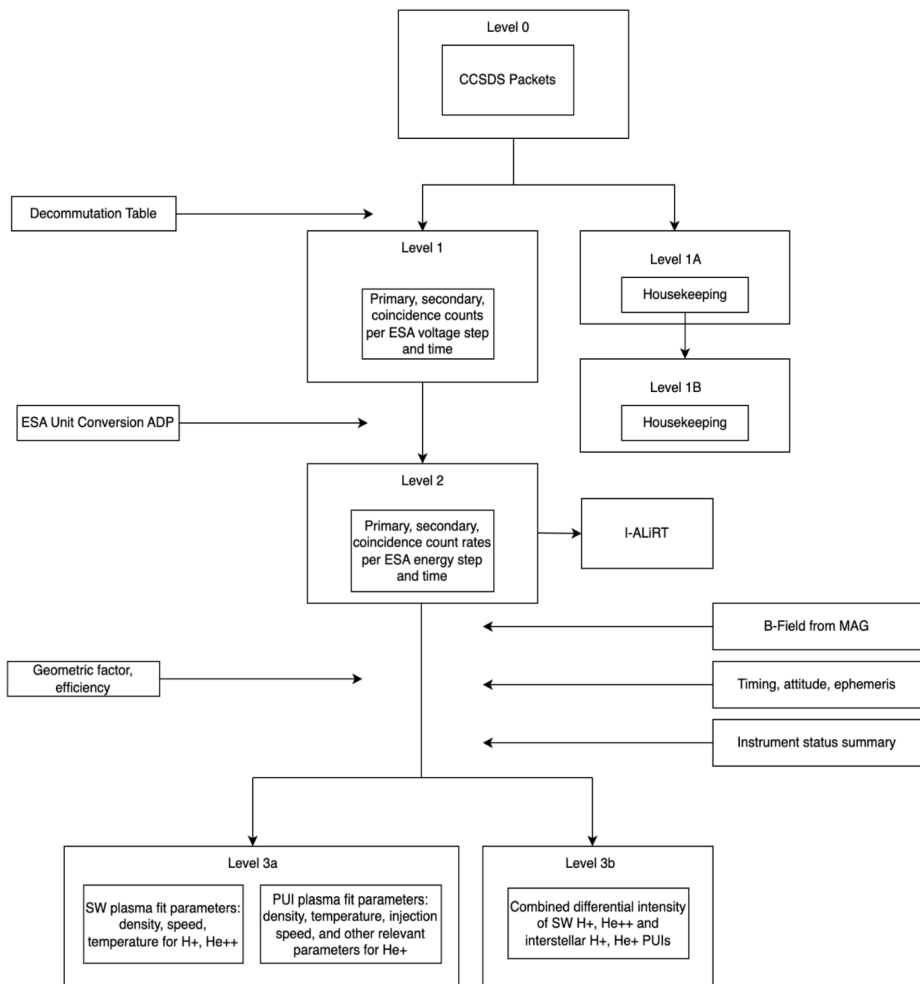


Fig. 32 IMAP SWAPI Data Pipeline Flow Diagram

At Level 1 and 2, counts are grouped by sweeps – each consisting of 72 energy steps taken over a 12 s interval. Packet boundaries are synchronized with the start of each sweep, with a total of 12 packets per sweep. Each sweep is transmitted as 12 one-second packets. Level 3 data products are delivered at a variety of cadences, combining multiple sweeps. The highest resolution baseline is 1 min for SW H⁺ and ⁴He⁺⁺ and 10 min for PUIs. Figure 32 summarizes each data product, including time resolution, components, inputs required, and associated ancillary files (see also Appendix C).

7.1 Level 0 to Level 1 Data Processing

Level 0 to Level 1 data processing decommutates the Level 0 CCSDS packets and extracts PCEM, SCEM, and coincidence counts. Only science telemetry packets are processed into Level 1 data products, although additional telemetry groups are referenced to support this processing. Housekeeping data from Level 0 are stored after converting DAC values into

physical using predefined conversion equations and coefficients for temperature readings, voltages and currents.

Level 0 data are flagged as Not a Number (NaN) during processing if: 1) the checksum is invalid, 2) the SWAPI mode is not HVSCI, 3) any of the count rates exceed 4.0 MHz, or 4) a CEM counter threshold has been exceeded and persists during the sweep. Additional data quality flags are set when thermistor temperatures, CEM currents, voltage tolerances, or count rates exceed predefined thresholds.

Uncertainty in the PCEM, SCEM, and coincidence counts is quantified during Level 1 processing. The dominant source of uncertainty is statistical and follows Poissonian behavior: $\Delta N = \sqrt{N}$, for count value N and uncertainty ΔN . An additional source of uncertainty arises from SWAPI's count compression mechanism (Sect. 6.4.5). This is characterized via modeling by synthetically generating high-range counts, applying compression and decompression, and comparing the results to derive an expression for estimating the compression-induced error.

7.2 Level 1 to Level 2 Data Processing

Level 1 to Level 2 processing converts raw counts into count rates and propagates associated uncertainties. For example, the PCEM count rate, C_{PCEM} , is obtained by dividing the counts (N_{PCEM}) by the live time for data acquisition ($t_{\text{bin}} = 0.145 \text{ s}$; Sect. 6.4.1): $C_{\text{PCEM}} = \frac{N_{\text{PCEM}}}{t_{\text{bin}}}$. These steps are also applied to the SCEM and COIN rates. The count rate uncertainty, ΔC , is just the Poissonian uncertainty divided by the time (e.g., $\Delta C = \frac{\sqrt{N}}{t_{\text{bin}}}$).

The Level 1 to Level 2 data processing also includes the conversion from ESA step numbers (from 0 to 71) to energy values (in eV) using the ESA Unit Conversion ADP LUT. No additional rejection, culling, or flagging of bad data is performed at this stage. However, a flag is added to data collected during the daily repointing maneuvers to indicate periods of SC thruster activity.

7.3 Level 2 to Level 3a Data Processing

The Level 2 to Level 3 processing is where the algorithms start to differentiate amongst the SW (H^+ , $^4\text{He}^{++}$) and PUI (He^+) populations. The Level 3a branch derives the plasma parameters (speed, density, and temperature) of the core SW protons (H^+), SW alphas ($^4\text{He}^{++}$), and PUI helium (He^+) from the count rates in the Level 2 product. To estimate these parameters, SWAPI's coarse scan histograms are modeled using the instrument response function and geometric factor, both derived from the combination of SIMION[®] simulations, Geant4 modeling, and laboratory calibration results. As described in Sect. 5, these components are compiled into a LUT, which accounts for dependencies on ESA energy and SW flow direction with respect to the SWAPI instrument axis (azimuth and elevation). The LUT is used to extract SW H^+ and $^4\text{He}^{++}$ parameters (Sects. 7.3.1 & 7.3.2) while PUI He^+ is derived using additional forward model fitting similar to the methodology employed on SWAP (McComas et al. 2025a and references therein; Sect. 7.3.3).

7.3.1 Solar Wind Protons (H^+)

SWAPI does not have built-in angular resolution and therefore the only way to differentiate between SW and PUIs is their energy per charge. However, for the case of SW protons, the directions can be resolved due to a statistically significant change in flux as a function of spin angle. SWAPI's 12 s ESA sweeps occur while the IMAP spacecraft spins every 15 s

about its Sun-facing axis. This results in a linear variation of the spin phase across the ESA sweep, causing the apparent location of the SW peak to shift over 5 consecutive sweeps.

The algorithm to address this first identifies the ESA step index with the highest count rates:

$$g_i = \min(\arg\max_{e \in K}(n_{i,e})), \quad (8)$$

$$h_i = \max(\arg\max_{e \in K}(n_{i,e})), \quad (9)$$

for lower bound, g_i , and upper bound, h_i , where $n_{i,e}$ is the number of counts observed in sweep i and ESA step, e , and K is the set of 62 ESA coarse steps.²

The center of the peak is then calculated using a weighted average over a window of ESA steps:

$$l_i = \frac{\sum_{e=g_i-\eta}^{e=h_i+\eta} e n_{i,e}}{\sum_{e=g_i-\eta}^{e=h_i+\eta} n_{i,e}} \quad (10)$$

where $\eta = 4$ defines a ± 4 -step window around the peak (blue shaded region in Fig. 29).

Once the peak center, l_i , is identified, the corresponding ESA energy (E_{i,l_i}) and spacecraft spin phase (φ_{i,l_i}) are interpolated:

$$E_{l_i} = E_{i,l_i} \quad (11)$$

$$\psi_{l_i} = \varphi_{i,l_i} \quad (12)$$

The interpolation is performed in logarithmic energy space to match the ESA step spacing. The uncertainty in peak position is derived from the Poissonian errors, propagated through the calculation of peak center.

To derive the SW bulk velocity parameters (flow speed and deflection angle), a trigonometric function fit is applied to the 5 consecutive (E_i, ψ_i) pairs, using the function:

$$E = A \sin(-\psi + \phi) + B \quad (13)$$

Fit parameters A , B , and ϕ are obtained via the Levenberg-Marquardt algorithm.

The bulk flow speed is calculated from the constant term B , for charge q and proton mass m_p :

$$u_{\text{SW}} = \sqrt{\frac{2Bq}{m_p}}. \quad (14)$$

The flow deflection angle θ is derived from the normalized amplitude A/B :

$$\theta = \zeta(u_{\text{SW}}) \frac{A}{B}. \quad (15)$$

Here, ζ is a constant dependent on SW conditions, determined via forward modeling of the SWAPI sweeps using OMNI data and stored in a LUT. The clock angle ϕ is the third

²In most cases we expect $g_i = h_i$, but if the highest and smallest ESA step index differ by more than 1 (i.e., $|g_i - h_i| > 1$), we have two separate maxima (which is unphysical) and therefore add a data quality flag.

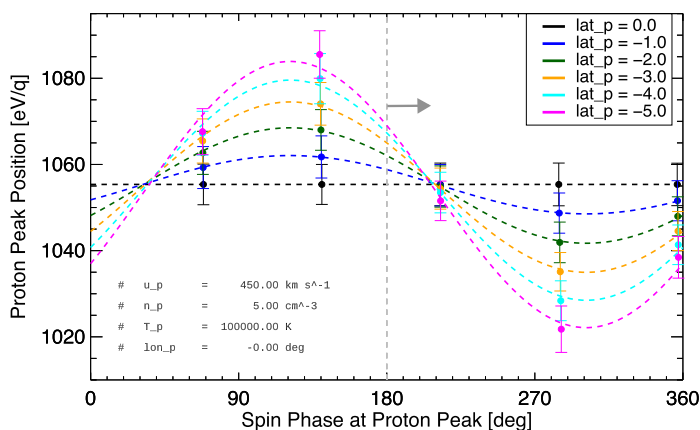


Fig. 33 Variation of the center position of the SW proton peak region with spin phase for six different flow deflection angles in elevation (lat_p from 0° to -5° ; below the instrument axis). The gray arrow highlights a systematic offset in the fitted peak position relative to the 180° spin phase (dashed vertical line). This offset is accounted for in the calibration LUT

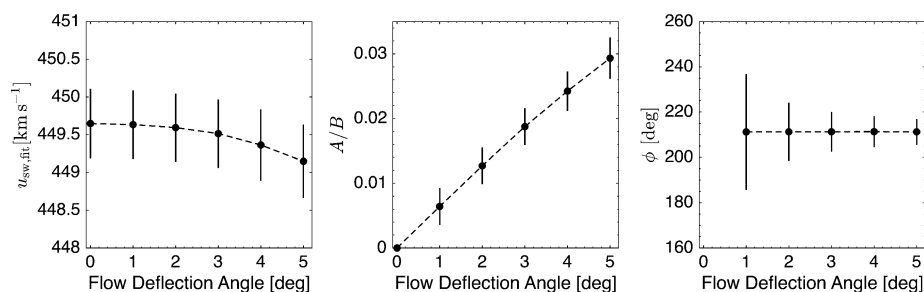


Fig. 34 Sine function fits applied to the SW H^+ peak positions as a function of flow deflection angles (absolute values) used to derive key plasma parameters, including: (left) estimated SW speed, (middle) variation of normalized amplitude (A/B), and (right) clock angle of solar wind deflection. Although the clock angle can have any value between 0° and 360° , this plot excludes 0° due to instability in the fitting routine. In all cases, the input clock angle is 180° , corresponding to flow deflection below the instrument axis. Error bars show the $1 - \sigma$ uncertainties

fit parameter derived from Eq. (13) and reflects the phase of the SW flow deflection with respect to the instrument axis in the IMAP de-spun frame (see Sect. 5.3). Uncertainties are propagated directly from the fits: flow speed uncertainty from B , deflection angle from both A and B , and clock angle from ϕ .

Figure 33 & 34 illustrate the application of this method. Figure 33 shows the sinusoidal variation of the SW proton peak with spacecraft spin phase for 6 different elevation angles of the flow deflection angles. Figure 34 (left panel) shows the accuracy of the bulk speed estimate, with fitted speeds matching the input speeds to within the $1 - \sigma$ uncertainty of the fitted value for most cases. There is also a slight systematic shift of $< 1 \text{ km s}^{-1}$ that increases with angle. The middle panel shows the linear relationship between normalized amplitude (A/B) and deflection angle, while the right panel displays the clock angle. The roughly 30° systematic offset evident here and in Fig. 33 varies with SW flow speed and is accounted for in the calibration LUT.

The SW H^+ density and temperature are derived using the analytical model of Elliott et al. (2016) adapted to incorporate SWAPI's instrument characteristics such as effective area ($A_{\text{eff}} = 1.633 \times 10^{-4} \text{ cm}^2$), SW azimuthal field of view ($\Delta\phi = 30^\circ$, corresponding to the angular extent of the sunglasses) and energy passband (E_c). For the SW sector, A_{eff} is defined as the ratio of SWAPI's aperture area (1.703 cm^2 ; Sect. 5.5) to the attenuation factor (1212; Sect. 4.1.2, Fig. 18) multiplied by the proton detection efficiency ($\varepsilon_{H^+} = 0.1162 \pm 0.0129$) measured in the lab.

The modeled count rate, $C(E_c)$, for each ESA energy passband is given by:

$$C(E_c) = \left(n_i A_{\text{eff}} \left(\frac{\beta}{\pi} \right)^{\frac{3}{2}} e^{-\beta(v_c^2 + u_i^2 - 2v_c u_i)} \right) \times \sqrt{\frac{\pi}{\beta u_i v_c}} \text{erf} \left(\sqrt{\beta u_i v_c} \left(\frac{\Delta\phi}{2} \right) \right) \left(v_c^4 \frac{\Delta v}{v_c} \sin^{-1}(v_{\text{th}}/v_c) \right), \quad (16)$$

where n_i is the SW density, u_i is the bulk flow speed, $v_c = \sqrt{2E_c/m_{H^+}}$ is the center speed of energy passband, Δv is the velocity width of the energy passband,³ $v_{\text{th}} = \sqrt{2k_B T_i/m_{H^+}}$ is the SW thermal velocity, $\beta = 1/v_{\text{th}}^2$, and 'erf' is the error function ($\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt$).

The density (n_i), and temperature (T_i) are treated as free parameters and are estimated by minimizing the inverse-variance-weighted squared difference between the modeled and observed count rates for each ESA sweep. This is done using a non-linear least square minimization algorithm (e.g., Levenberg-Marquardt algorithm). The 1-min resolution data product is achieved by averaging over 5 sweeps.

The analytical model tends to produce systematic offsets in the derived density and temperature, depending on SW conditions (e.g. see Fig. 29 in Elliott et al. 2016). These offsets are corrected using a calibration LUT precomputed from a wide range of SW scenarios based on the OMNI data set. The corrected values are given by:

$$n_{\text{sw}} = L(u_{\text{sw}}, \Theta, \Phi) E[n_i] \quad (17)$$

$$T_{\text{sw}} = K(u_{\text{sw}}, \Theta, \Phi, n_{\text{sw}}) E[T_i], \quad (18)$$

where $E[n_i]$ and $E[T_i]$ are the mean values from the five-sweep fit and L and K are correction factors from the LUT as functions of flow speed and direction.

7.3.2 Solar Wind Alphas ($^4\text{He}^{++}$)

SWAPI's SW alpha ($^4\text{He}^{++}$) parameters include flow speed, density, and temperature, returned at a 1 min cadence. Unlike protons, the alpha flow direction cannot be independently derived due to statistical limitations of the data, so we assume that it is the same as the proton flow direction. The bulk flow speed is obtained using a methodology similar to that of the protons (Sect. 7.3.1), but with minor differences: (1) instead of fitting individual sweeps, the analysis is performed by averaging over 5 sweeps. As a result, the sweep-specific indices (i) are dropped from Eqs. (8) to (12). (2) The "SW alpha region" is defined as the energy bin corresponding to the alpha peak, plus two bins on either side so that $\eta = 2$ in Eq. (10) (see red shaded region in Fig. 29). (3) no trigonometric fits are applied to derive the bulk

³Derived from the energy resolution, via: $\frac{\Delta v}{v} = \frac{1}{2} \left(\frac{\Delta E}{E} \right)$. The energy resolution, $\left(\frac{\Delta E}{E} \right)$, is obtained by taking the FWHM of a constant-angle sweep of the passband.

flow velocity parameters for alphas. Instead, only the bulk speed is calculated (no deflection angle) from the interpolated energy at the peak center (E_l) of the alpha region:

$$u_{\text{He}^{++}} = \sqrt{\frac{2E_l q_{\text{He}^{++}}}{m_{\text{He}^{++}}}} \quad (19)$$

where $q_{\text{He}^{++}}$ and $m_{\text{He}^{++}}$ are the ${}^4\text{He}^{++}$ charge and mass, respectively.

Temperatures and densities of the SW alphas are derived using the same analytical model used for the protons (Eq. (16)), again with appropriate substitutions for mass and charge ($v_e = \sqrt{2E_e q_{\text{He}^{++}}/m_{\text{He}^{++}}}$ and $v_{\text{th}} = \sqrt{2k_B T/m_{\text{He}^{++}}}$).

Finally, the analytical model introduces systematic offsets in the derived alpha parameters too. These are also corrected using the calibration LUT, with values given by:

$$n_{\text{He}^{++}} = L(u_{\text{He}^{++}})n, \quad (20)$$

$$T_{\text{He}^{++}} = K(u_{\text{He}^{++}}, n_{\text{He}^{++}})T, \quad (21)$$

where n and T are the density and temperature from the analytical model fit and L and K are correction factors derived from the LUT.

7.3.3 Pickup Ion Helium (He^+)

SWAPI's He^+ PUI parameters are derived by fitting statistically combined count rates over a 10-min cadence (50×12 s sweeps), for energies above the theoretical H^+ PUI cutoff (purple shaded region in Fig. 29). The fitting uses a forward model adapted from SWAP processing (following Chen et al. 2014; McComas et al. 2017, 2021, 2025a), with free parameters including cutoff speed, ionization rate, cooling index, and background rate to yield quantities such as PUI density, temperature, and injection speed. Velocities are transformed from the SWAPI frame to the SW frame using:

$$\mathbf{v} = (\mathbf{v}' + \mathbf{V}_{\text{IMAP}}) - \mathbf{u}_{\text{SW}} \quad (22)$$

where $\mathbf{v}'$ is the PUI velocity in the GSE frame, $\mathbf{V}_{\text{IMAP}}$ is the spacecraft velocity (GSE), and $\mathbf{u}_{\text{SW}}$ is the average SW velocity over the 10-min interval (see Sect. 7.3.1). Interstellar neutral helium density is computed using the hot model (Thomas 1978) over a range of distances from the Sun and different positions of spacecraft relative to the interstellar flow. These densities are precomputed and stored in a LUT.

Modeled count rates (C_m) are obtained by integrating the distribution function over the instrument's energy and angular response (see Eq. (6) of Sect. 5.3) and converted to counts (N_m) via: $N_m(E_e) = 50 * t_{\text{bin}} * C_m(E_e)$ where $t_{\text{bin}} = 0.145$ s (see Sect. 7.2). Only energy bins above the H^+ PUI cutoff ($E_e > E_{b, H^+}$) are used, excluding bins close to the background count rate (e.g., ~ 0.1 Hz in Fig. 29). The cutoff energy is: $E_{b, H^+} \approx \frac{1}{2} m_{H^+} \left(2v'_{b, H^+} \right)^2$, for $v'_{b, H^+} = |\mathbf{u}_{\text{SW}} - \mathbf{u}_H - \mathbf{u}_{\text{IMAP}}|$, with $\mathbf{u}_H = 22 \text{ km s}^{-1}$ coming from $(252.2^\circ, 9.0^\circ)$ in *ecliptic J2000* coordinates (Lallement et al. 2005) and the spacecraft velocity, $\mathbf{u}_{\text{IMAP}}$, now in the solar inertial frame.

The best-fit model minimizes the Poisson likelihood chi-square (Cash 1979; Baker and Cousins 1984; Ji et al. 2020):

$$\chi^2_{\lambda, p} = 2 \sum_e \left(N'_m(E_e) - N(E_e) + N(E_e) \ln \left(\frac{N(E_e)}{N'_m(E_e)} \right) \right), \quad (23)$$

where $N'_m(E_e) = N_{bkg} + N_m(E_e)$ includes background counts.

He^+ PUI density and temperatures computed via (Piessens et al. 1983):

$$n_{\text{He}^+ \text{ PUI}} = \int_0^{v_b} \int_0^\pi \int_0^{2\pi} f(v) v^2 dv \sin \theta d\theta d\phi = 4\pi \int_0^{v_b} f(v) v^2 dv \quad (24)$$

$$T_{\text{He}^+ \text{ PUI}} = \frac{m_{\text{He}^+} \int_0^{v_b} f(v) v^4 dv}{3k_B \int_0^{v_b} f(v) v^2 dv} \quad (25)$$

Uncertainties in n_{He^+} and T_{He^+} are estimated by manually propagating 1σ uncertainties in the fit parameters:

$$\sigma_{n_{\text{He}^+ \text{ PUI}}} = \sqrt{\sum_{j=1}^4 \left(n_{\text{He}^+ \text{ PUI}} - (n_{\text{He}^+ \text{ PUI}})_{j+\sigma_j} \right)^2} \quad (26)$$

$$\sigma_{T_{\text{He}^+ \text{ PUI}}} = \sqrt{\sum_{j=1}^4 \left(T_{\text{He}^+ \text{ PUI}} - (T_{\text{He}^+ \text{ PUI}})_{j+\sigma_j} \right)^2} \quad (27)$$

Automated flags identify some rare events where the SW flow deflection angle exceeds 5° or the bulk speed falls outside the LUT calibration range (e.g., $u < 300 \text{ km s}^{-1}$ or $u > 1000 \text{ km s}^{-1}$).

7.4 Level 2 to Level 3b Data Processing

Level 3b processing converts SWAPI's Level 2 count rates into combined differential intensities for SW (H^+ and $^4\text{He}^{++}$), and interstellar PUIs (H^+ and He^+) via the relation:

$$J\left(\frac{E}{q}\right) = \frac{C(E/q)}{\frac{E}{q} * G(E/q) * \varepsilon}, \quad (28)$$

where $J\left(\frac{E}{q}\right)$ is the differential intensity in units of $\#/\text{[cm}^2 \text{ s sr eV/q]}$, $C(E/q)$ is the count rate, $G(E/q)$ is the geometric factor, and ε is the detection efficiency, which is updated periodically via gain tests and stored in a LUT.

The uncertainty in differential intensity is propagated from uncertainties in count rate (ΔC) and energy per charge, $\Delta(E/q)$:

$$\Delta J = J \sqrt{\left(\frac{\Delta C}{C}\right)^2 + \left(\frac{\Delta(E/q)}{(E/q)}\right)^2}. \quad (29)$$

8 Conclusions

The SWAPI instrument onboard IMAP is the first NASA flight instrument to be designed, built, calibrated and delivered out of the Space Physics Laboratory at Princeton University. Its architecture builds on the heritage of the SWAP instrument, with enhancements optimized for continuous operation at 1 au. A key innovation is the aperture grid assembly, which attenuates the solar wind flux by approximately three orders of magnitude while preserving a large geometric factor for pickup ions. This configuration enables simultaneous measurements of these populations at high time cadences within a single, compact instrument.

SWAPI meets or exceeds all science and performance requirements defined at the outset of the IMAP mission. Ground-based calibration, cross-instrument comparisons, and in-flight gain tests demonstrate the instrument's reliability and accuracy across its full operational range. The instrument's dual coincidence detector system provides robust background suppression and qualitative species discrimination, while its high-voltage electrostatic analyzer enables fine energy resolution across a broad dynamic range.

As part of the IMAP mission, SWAPI's measurements contribute to a comprehensive understanding of the heliosphere as a coupled system. Through coordinated observations of the solar wind and interstellar pickup ions at high cadence and resolution, SWAPI strengthens the observational foundation needed to address fundamental questions about solar wind–interstellar medium interactions, particle energization, and heliospheric structure. As the IMAP mission advances, SWAPI's early measurements are poised to play a central role in achieving the mission's transformative science goals, including determining the flow properties of the local interstellar medium and characterizing particle acceleration processes throughout the heliosphere.

Appendix A: SWAPI Science Requirements

All of SWAPI's science requirements flow from the following two IMAP mission requirements:

1. MRD-L2-16: Thermal Ions

- The Mission shall measure thermal (solar wind) ions, as follows:
 - Energy range: 400 eV/q - 10 keV/q
 - Energy Resolution $\Delta E/E$: Better than 10%
 - Energy bands per decade: ≥ 5
 - Ion species: H^+ and $4He^{++}$
 - Cadence: ≤ 1 min for 1D VDFs of H^+ and $4He^+$
- Traced directly from IMAP's Level 1 baseline science requirement 4.1.1.8 (NASA-L1-77).

2. MRD-L2-210: Pickup H^+ and He^+

- The Mission shall measure the combined 1D VDFs of H^+ and He^+ , as follows:
 - Energy range: 0.7 - 18 keV/q
 - Energy bands: ≥ 5 per decade
 - Cadence: ≤ 10 minutes
- Traced directly from IMAP's Level 1 baseline science requirement 4.1.1.11 (NASA-L1-82).

SWAPI's ability to meet its science requirements has been verified through the combination of testing and analysis, summarized in Table 12. Results demonstrate that SWAPI will meet, and in most cases, exceed these in its expected performance.

Appendix B: Princeton Space Physics Calibration Facility

The Space Physics Laboratory at Princeton University is an experimental laboratory dedicated to the prototyping, building, testing, and calibration of sensors and instrumentation

Table 12 Summary of SWAPI's science requirements and verification results

Measurement	SWAPI ID	Requirement	Verification Method	Expected Performance
Solar Wind & Pickup-Ions	RSWAPI-1046	SWAPI shall measure Solar Wind (SW) H^+ and $^4He^{++}$ along with interstellar He^+ and H^+ Pickup Ions (PUIs).	Forward model, Calibration	Full Compliance
Solar Wind H^+	RSWAPI-1048	SWAPI energy range for measuring the solar wind H^+ shall be 0.4-10 keV/q.	Calibration	Exceeds (0.089 to 21.4 keV/q)
	RSWAPI-1049	SWAPI energy resolution (dE/E) for measuring the solar wind protons H^+ shall be better than 10 percent.	Calibration	Exceeds (8.1%)
	RSWAPI-1050	SWAPI energy bands per decade for measuring the solar wind H^+ shall be greater than or equal to 5.	Calibration	Exceeds (~27 energy bands per decade)
	RSWAPI-1051	SWAPI cadence for measuring the solar wind H^+ shall be less than or equal to 1 min for 1D VDF.	Forward model, Calibration	Full Compliance
Solar Wind $^4He^{++}$	RSWAPI-1053	SWAPI energy range for measuring the solar wind $^4He^{++}$ shall be 0.4-10 keV/q.	Calibration	Exceeds (0.1 to 20 keV/q)
	RSWAPI-1054	SWAPI energy resolution (dE/E) for measuring the solar wind $^4He^{++}$ shall be better than 10 percent.	Calibration	Exceeds (8.1%)
	RSWAPI-1055	SWAPI energy bands per decade for measuring the solar wind $^4He^{++}$ shall be greater than or equal to 5.	Calibration	Exceeds (27 energy bands per decade)
	RSWAPI-1056	SWAPI cadence for measuring the solar wind $^4He^{++}$ shall be less than or equal to 1 min for 1D VDF.	Forward model, Calibration	Full compliance
Pickup-Ions	RSWAPI-1061	SWAPI shall measure the combined interstellar 1D VDFs of He^+ and H^+ Pickup Ions (PUIs).	Forward model	Full Compliance
	RSWAPI-1062	SWAPI energy range for measuring He^+ and H^+ PUIs shall be 0.7 keV/q - 18 keV/q.	Forward model, Calibration	Exceeds (0.089 to 21.4 keV/q)
	RSWAPI-1063	SWAPI energy bands per decade for measuring He^+ and H^+ PUIs shall be greater than or equal to 5.	Forward model, Calibration	Exceeds (27 energy bands per decade)
	RSWAPI-1064	SWAPI cadence for measuring the combined He^+ and H^+ PUIs shall be less than or equal to 10 min	Forward model, Calibration	Full Compliance
Geometric Factors	RSWAPI-1060	SWAPI effective geometric factor (at 1 keV/q) for SW Sun-viewing shall be greater than or equal to $1 \times 10^{-8} \text{ cm}^2 \text{ sr eV/eV}$ for SW H^+ .	Calibration, Forward model, Analysis	Exceeds ($1.00 \times 10^{-7} \text{ cm}^2 \text{ sr eV/eV}$)
	RSWAPI-1065	SWAPI effective geometric factor (at 1 keV/q) for PUI He^+ non-Sun-viewing shall be greater than or equal to $8 \times 10^{-3} \text{ cm}^2 \text{ sr eV/eV}$.	Calibration, Forward model, Analysis	Exceeds ($2.25 \times 10^{-2} \text{ cm}^2 \text{ sr eV/eV}$)
Rates	RSWAPI-1058	SWAPI shall have a dynamic range of 300 kHz to 0.3 Hz counting rates.	Analysis, Component Test, Calibration	Full Compliance
	RSWAPI-1059	SWAPI SNR shall be greater than or equal to 1 at 0.3 Hz coincident count rate.	Calibration	Exceeds (>10)

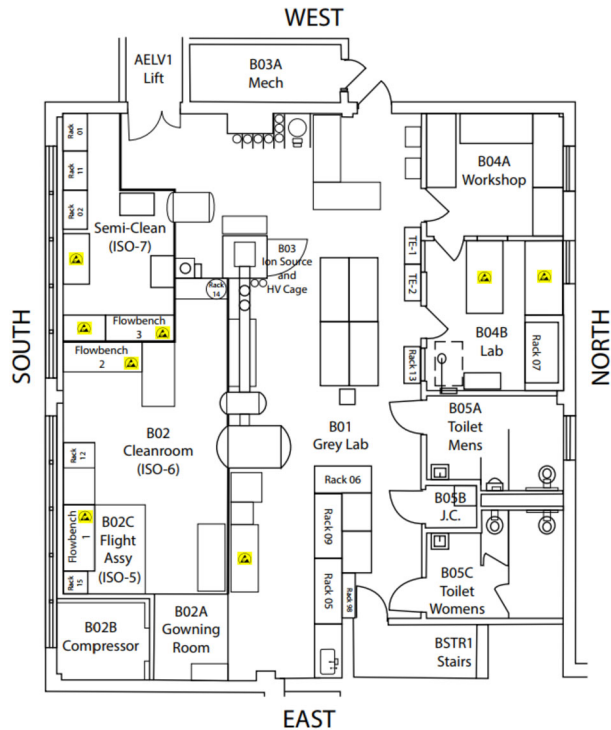
for space applications, including NASA space-flight hardware, while also serving as a teaching laboratory. In addition to hosting Junior Projects, Senior Theses, and Masters and PhD projects, Princeton University undergraduates from all majors and at all levels have an opportunity to gain hands-on, real-world experience through a two-part laboratory course (listed as AST 250 & AST 251). The first semester focuses on teaching practical skills, such as running procedures, operating ultra-high vacuum equipment, working in a cleanroom, handling various types of tools, performing basic mechanical and electrical measurements, collecting and analyzing experimental data, etc. and culminates in a Preliminary Design Review, in which students present the preliminary design of a significant project of their team's choice. The second semester is then dedicated to the implementation of the group-based research project, including design, manufacturing, assembly, and test phases, with the final assessment performed through a Critical Design Review.

The Space Physics at Princeton group (<https://spacephysics.princeton.edu/>) was founded in March 2016 when Prof. McComas moved to Princeton. The group grew quickly and over its first three years, and in this time, identified, remodeled, and moved into a three-story, 6000+ sq ft wing of a former elementary school building, which had most recently served as a preschool. The Space Physics Laboratory occupies the bottom floor of this building. On July 1, 2020, during the COVID-19 pandemic, a small subset of the team began to build out the technical infrastructure and equipment of the lab. Three of IMAP's flight instruments were tested and calibrated in this facility: SWAPI, from September 16 to October 2, 2024, IMAP-Lo (Schwadron et al. 2025) from October 15 to 26, 2024, and the combination of SWAPI and CoDICE for cross calibration in the same beam (Sect. 4.2; Livi et al. 2025) from June 2 to June 10, 2025.

The layout of the >2000 sq ft laboratory is configured to optimize the workflow from receiving and cleaning of parts to assembly and inspection to testing and calibration, each in highly controlled environments to ensure NASA workmanship and cleanliness standards. The schematic of Fig. 35 provides an overview of the various sections of the lab, including the main "grey" lab (929 sq ft), workshop (138 sq ft), cleaning room (172 sq ft), gowning room (57 sq ft), main and semi-cleanrooms (555 sq ft), ion source cage (17 sq ft), and Calibration System. The cleanrooms consist of soft-wall, vertical flow-type regions partitioned into two parts: 1) the Main Cleanroom, which is certified as ISO class 6 (<1000 particles/cu ft) and 2) a Semi-cleanroom (ISO class 7; <10,000 particles/cu ft). All flight hardware assembly occurs in a separately partitioned flow bench region in the Main Cleanroom (ISO Class 5; <100 particles/cu ft). The soft-wall cleanroom is advantageous for retaining cleanliness; cutouts were made for each of the chamber doors to open into the cleanrooms, ensuring minimal contamination during instrument assembly and installation. Ultra-pure nitrogen (99.999%) is supplied by an in-house generator located on the West wall of the lab, with backup bottles in reserve.

An example of the cleanliness workflow is provided here: received parts are hand-cleaned with IPA in the cleaning hood (sometimes preceded by other cleaners like Alconox at the sink). Small parts additionally undergo cleaning in the ultrasonic cleaner. Vacuum-compatible parts are then transferred into a Bakeout Chamber (30" outer diameter, 40" long; Fig. 36) located on the far East side of the lab (Fig. 35). Temperatures on the system are controlled via heaters attached to a large copper platen. Two temperature controllers provide a feedback control loop, and temperature is monitored by three sensors attached to RTD's that reside inside the chamber. The platen is enclosed by a folded plate of aluminum housing wrapped in a multi-layer aluminum shroud, which provides reasonable thermal insulation and temperature uniformity throughout the central volume. The door of the Bakeout Chamber opens into the semi-cleanroom via a cutout in the soft-walled curtain. Parts are transferred by personnel wearing semi-cleanroom attire (lab coats, boot covers, beard covers,

Fig. 35 Layout of the Princeton Space Physics Laboratory



gloves, face masks, and hairnets) and installed in the chamber. The Bakeout Chamber is pumped by a water-cooled magnetically levitated turbo backed by an oil free rough pump. Parts are then baked out at pre-defined temperatures and high vacuum pressures (nominally low 10^{-7} torr) until the Residual Gas Analyzer (RGA; 200 AMU with CEM) scan comes out clean (pressure dominated by atmospheric gases, with heavy elements > 60 at nominally $< 10^{-12}$ torr). Once a part comes out of bakeout, it can be inspected and/or transferred directly into the Main Cleanroom through another passthrough cut into the soft-walled curtain that separates the two and transferred to personnel in full cleanroom attire to be used in assembly or directly installed into the Main Chamber or Science Chamber for testing.

The Space Physics Calibration System is located in the central part of the lab and consists of a Main Chamber (48" outer diameter; 40" deep), Science Chamber (24" outer diameter, 36" deep), Flight Tube (6" outer diameter, 111" long), charge-exchange vessel (6" four-way cross), and an ion source region (19" long), each separated by 6" gate valves as depicted in Figs. 37 & 38. The gate valves are arranged to separate each of the subsections, each of which are independently pumped, either by magnetically levitated turbos (550 L/s; 48,000 rpm) or cryogenic pumps (1500 L/s; mounted on 8" gate valves) and all backed by separate 15 L/s oil-free rough pumps. These subsystems can operate independently and can be valved off and brought to atmosphere while maintaining high vacuum throughout other parts of the system. In addition, while tests on student projects are being performed in the Science Chamber, the Main Chamber can be kept isolated to house flight-ready instruments or sensors under vacuum and ready to test.

Nominal operating pressures in the Main Chamber and Science Chamber (without flight instruments installed) are of the order of $\sim 10^{-8}$ to 10^{-9} torr, with RGAs typically showing no high mass contaminants (> 60 AMU) above $\sim 10^{-12}$ Torr (Fig. 39). Beam imaging

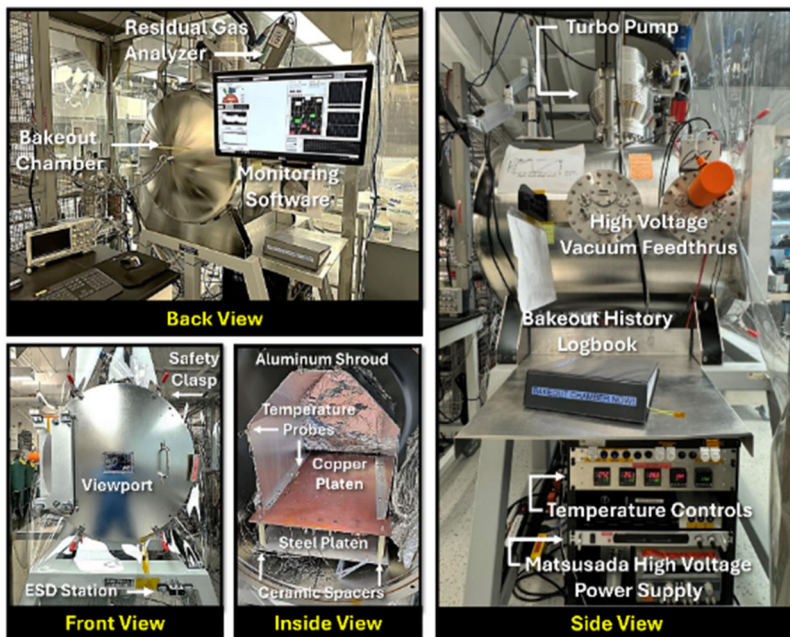


Fig. 36 Multiple perspectives of the Bakeout Chamber system used to clean and bakeout flight hardware prior and other elements prior to assembly or testing

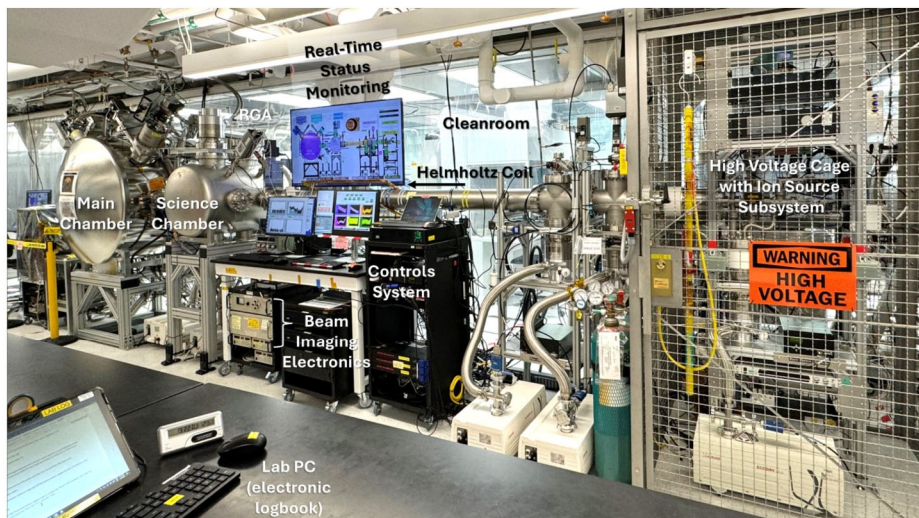


Fig. 37 Image of the Princeton Space Physics Calibration System, used for flight instrument development and calibration of both SWAPI and IMAP-Lo, as well as the cross-calibration of SWAPI and CoDICE

sensors, including 2 absolute beam monitors (ABMs; Funsten et al. 2005) and a 40 mm active-area imaging microchannel plate (MCP) are also housed in the two chambers and were used during calibration and cross-calibration to monitor the spatial and intensity dis-

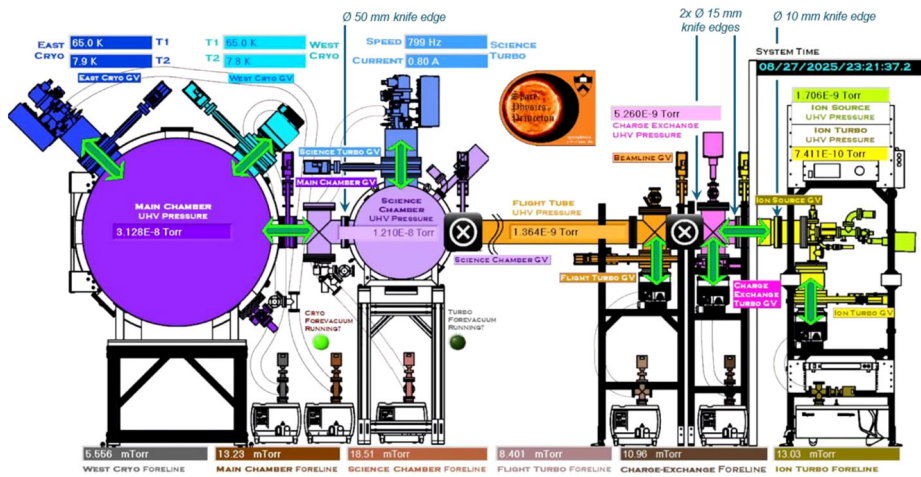


Fig. 38 Screenshot of the real-time monitoring “Big Board” display of the Princeton Space Physics Calibration System, in the same configuration as used for SWAPI-CoDICE cross calibration in June 2025. Variations of this configuration were used for SWAPI and IMAP-Lo FM calibrations as well

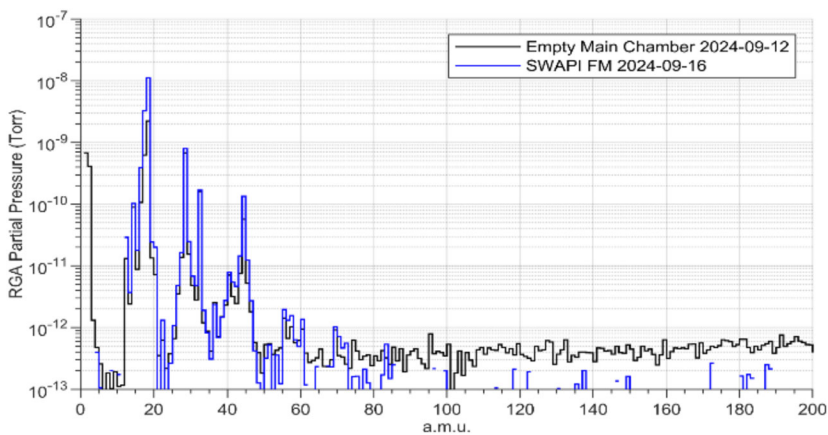


Fig. 39 RGA scans showing the Main Chamber baseline before (black) and after (blue) SWAPI’s installation for its flight calibration campaign in September 2024. In both cases, the Main Chamber also contained 5 stages, a 40-mm active area imaging MCP, 2 ABMs, and a UV source

tributions of the beam (Fig. 40). These are mounted on a 2-axis system – designed via an undergraduate Senior Thesis student project – with a “paddle” system affixed to a rotational stage and mounted to translational stage to allow for selection of up to 8 paddles, apertures, or sensors in the beam and flexibility in mapping over the entire beam area.

During SWAPI’s engineering and flight calibrations, the instrument was mounted on a 3-axis positioning system consisting of a “Horizontal” linear translational stage (± 100 cm travel) and two rotational stages (“Outer” for yaw, “Inner” for roll; $\pm 170^\circ$ rotation) mounted on top of each other, as shown in Figs. 41 & 42 (see also Sect. 4.1). This same 3-axis approach was also used for IMAP-Lo’s calibration but attached to different mechanical GSE (Schwadron et al. 2025). Finally, a 2-axis configuration was adopted for the cross-calibration

Fig. 40 Photo of the beam monitoring system in the Princeton Main Chamber. Depicted here, CCW from 12 o'clock, are the Lo-CEM, the MCP, a collimator paddle, and the carbon-foil ABM

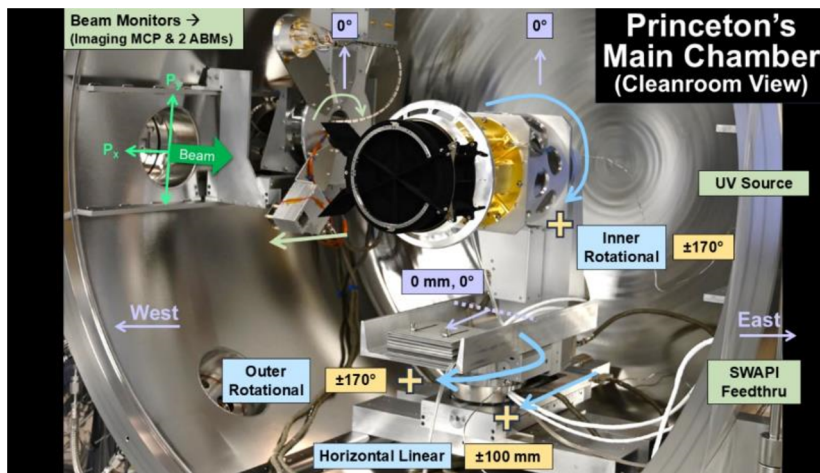


Fig. 41 Cleanroom view of Princeton's Main Chamber with the SWAPI flight instrument installed, depicting configuration during FM-calibration, including the 3-axis positioning system, beam monitors, SWAPI feedthroughs, UV source, and incoming direction of the beam

of SWAPI & CoDICE, as shown in Fig. 43 and described in Sect. 4.2. In all cases, the fixture was mounted so that cables could be fed through the center of the Outer Rotational stage and therefore avoid complications from extra motion of cables. A camera mounted on top of the Main Chamber enables continual monitoring while performing stage motions via customized software designed specifically for use in the Princeton Lab. All GSE used to mount instruments and sensors is engineered to fundamentally avoid collisions, but safety limits are also incorporated into the software to add flexibility for play in cables and ensure physical margin during instrument operations and testing.

The total length of the beamline from the exit of the ion source to the entry flange on the Main Chamber is 4.62 m. Knife-edges located along the beamline (shown in Fig. 38)

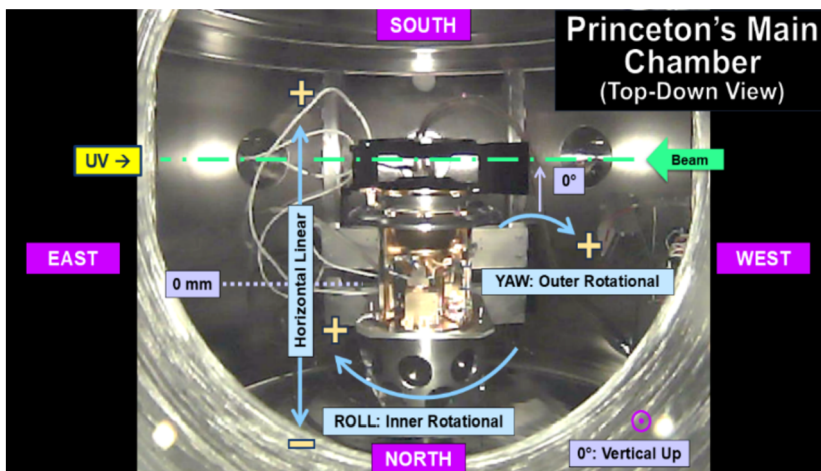


Fig. 42 Top-down view of the SWAPI flight instrument installed in the Princeton Main Chamber, here operating under vacuum. Angular and translational directions are labeled, along with the rough cardinal directions within the Princeton Space Physics Lab

are used to constrain the beam so that by the time it reaches the entry port on the Main Chamber, it is highly uniform, parallel, and expanded to a diameter of 6.25 cm (7.05 cm at the entrance of SWAPI) but can be reduced to any reasonable smaller size by positioning paddle-mounted collimators in the beam.

The Princeton ion source, illustrated in Fig. 44, is comprised of an electron impact ionization source complete with an integrated Wien filter. The source itself is a two-lens, extractor type, focused, differentially pumped ion gun which can raster a 10 mm x 10 mm area of the surface at the pre-determined working distance. The ion source gas supply consists of a neutral gas mixture of Hydrogen, Deuterium, Helium, Neon, and Argon balanced by Nitrogen. The flow rate into the ion source is controlled by a precision leak valve. The pressure in the ion source filament region is linearly correlated with beam intensity; thus, this precision flow control translates to a highly stable beam (intensity variations of $\lesssim 10\%$ over day-long timescales at a single fixed setting). A Wien filter built into the ion source optics is used to distinguish amongst the various species, thus enabling selections of Proton (H^+), Deuterium (D^+), Hydrogen (H_2^+), Deuterium Hydrate (DH^+), Helium (He^+), Deuterium (D_2^+), Nitrogen (N^+ , N_2^+), Neon (Ne_{20}^+ , Ne_{22}^+), Argon (Ar^+), or water group (O^+ , OH^+ , H_2O^+) ion beams with desired beam spatial width, energy, and count rate. The Princeton Calibration System was designed to generate stable ion beams with energies ranging from <100 eV to ~ 30 keV. The ≤ 5 keV beams are produced by the ion source when it is at neutral ground (0 V), while the higher energy range (5 keV to 30 keV) is achieved by floating the ground on a high voltage power supply with a positive-polarity module to generate a post-acceleration energy based on the sum of the source energy and the difference in potential between the floated and neutral ground.

Immediately downstream of the high voltage cage, between the ion source and beamline gate valves resides the “charge exchange” cell (colored pink in Fig. 38), that expands the beamline capabilities to include neutral beams. This region consists of a roughly foot-long ConFlat tee mounted to a turbo at below, and a precision leak valve on top. Supplied by an ultra-high purity (99.999%) nitrogen gas bottle, the precision leak enables precise control of gas flow into the chamber. For safety control, there is a predefined maximum number of

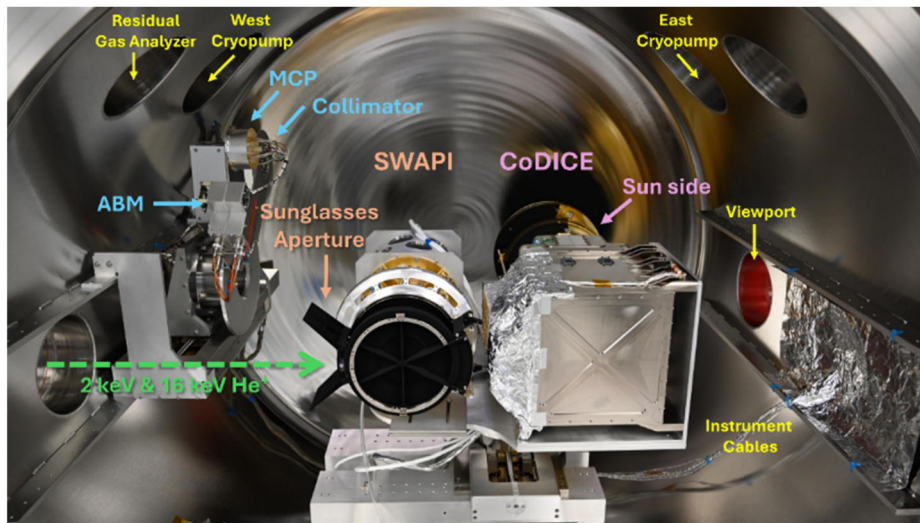
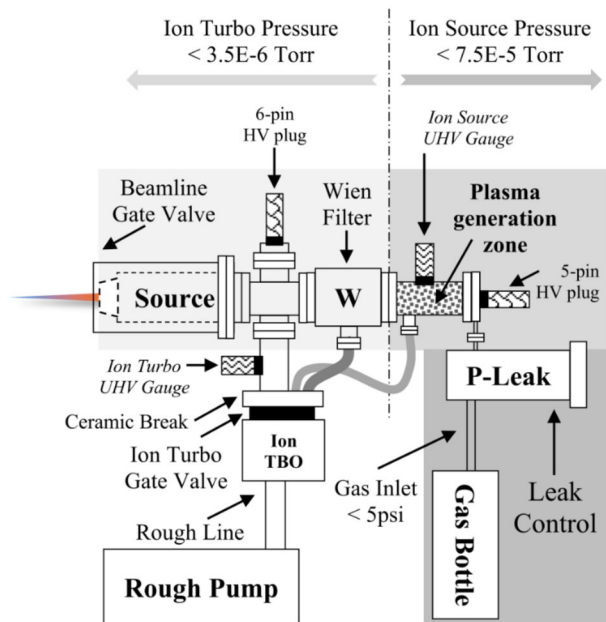


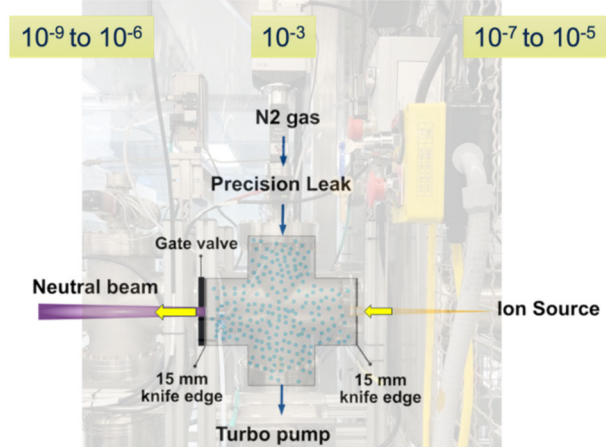
Fig. 43 Cleanroom view of Princeton's Main Chamber with the SWAPI and CoDICE flight instruments installed in the two-axis configuration use for cross-calibration. The instruments here are oriented to target the solar wind azimuthal sector (e.g., beam through SWAPI's sunglasses)

Fig. 44 Simplified diagram of the ion source section, including its pumps and gas feeding system. Operational pressures prescribed by the ion source manual are denoted by the differing shades of grey (top left and right; also see arrows). Note that UHV and HV are the abbreviation of ultra-high vacuum and high voltage, respectively



steps that a user can dial in. Much like the charge exchange processes that occur in the outer heliosphere (e.g., McComas et al. 2025b; Reisenfeld et al. 2025). Figure 45 illustrates the process, with ions from the source region (right) entering the charge exchange region (middle) and exiting through the flight tube as a fraction of neutrals (for example, experiments in the lab have shown that around $\sim 10\%$ of H^+ ions at 1 keV are neutralized, consistent

Fig. 45 Schematic illustrating the charge-exchange process in the Princeton beamline, by which ions get converted to neutrals as they pass through the charge-exchange region filled with N₂ gas, kept at elevated pressures compared to the surrounding beamline (here labeled in units of Torr). The region is bounded on either side by two 15 mm knife edges which enables differential pumping at sufficient pressures to maintain a reasonable rate of charge exchange



with Lindsay and Stebbings 2005). The remaining ions are then diverted downstream by a Helmholtz coil. Low-energy neutral beams from this system were used extensively during IMAP-Lo's calibration campaigns and is further described by Schwadron et al. (2025).

Roughly a meter beyond the charge exchange section is a specialized Helmholtz coil (mounted to the flight tube) that serves to compensate (rather than exclude) for the effects of the Earth's magnetic field (Khoo et al. 2023). Early testing in the Princeton beamline demonstrated the geomagnetic correction to be especially relevant for < 5 keV protons and < 1 keV Helium or equivalent. When switching between SWAPI, IMAP-Lo or beam-monitor targets, slight adjustments in the Helmholtz location were made to account for a given instrument or sensor's position with respect to the beam's entrance into the Main Chamber. Proper setup of the Helmholtz coil was essential for achieving low-energy measurements in SWAPI (down to ~ 100 eV) and IMAP-Lo (Schwadron et al. 2025), enabled complete mapping for the low-energy passbands (e.g., 200 eV H⁺), and ensured uniformity when performing the continuous energy sweep tests (Sects. 4.1.1 & 4.1.2). At the lowest energies, cross checks with Argon were performed for each type of test, demonstrating nearly identical angular and energy responses to helium and proton beam counterparts.

The architecture of the lab, from the choice of vacuum hardware to the operations of the control software, to the ESD floors, chairs, lab coats, and flow benches and workstations, to the cleanliness procedures and processes were carefully designed with instrument safety and optimization in mind. In the Calibration System, critical components – including gauges, valves, pumps, and the ion source subsystem – are controlled via dedicated panels by the “Controls PC”, which allows for only one-way communications and is “air gapped” to ensure only local control. Safeties are built into the software to close gate valves and isolate sections from the rest of the system if errors are encountered in the ion source region, or if pressures get too high. Gate valves on the Main Chamber automatically close if the pumps experience errors or their temperatures get too high, and an HV interlock kicks in to power off any high voltages if there is power loss to the system or pump valves suddenly close. The software generally restricts users from performing unsafe actions (e.g., gate valve control is disabled when it is unsafe to open a vented chamber to a section under vacuum or a spinning turbo), but in no cases does the software cause valves to automatically open.

In the case of power outage, the essential parts of the lab are kept on a backup UPS, which provides ~ 30 min of backup power – sufficient time to perform a controlled shutdown under severe circumstances or ride through shorter power anomalies without interruption.

Even when power is lost to the entire system, all valves automatically close, the turbos perform controlled spin-downs, and other pumps power off. The entire system is additionally designed to record and provide real-time status updates, including valve states, pressures, cryopump temperatures (on the Main Chamber and Science Chamber), turbo pump speeds (beamline and ion source sections), positioning system locations (Main Chamber and Science Chamber), and ion source settings. This data is recorded at a 1 s cadence and stored in a locally hosted database. Details about the state of the entire system, along with the data acquired from the RGAs and beam imagers were tracked throughout SWAPI's instrument development, including during its final calibration, during IMAP-Lo's development and calibration for the parts which took place at Princeton and for the SWAPI-CoDICE cross-calibration campaign. Full data sets of the above-mentioned variables, along with details of the as-built system hardware and software configurations have been collected and archived along with the instrument data. The real-time state of the Space Physics at Princeton Laboratory can generally be seen online at: <https://spacephysics.princeton.edu/cal-system-status>.

Appendix C: Summary of SWAPI Science Data Products

The list below details the time resolution, components, inputs required, and ancillary files for each of the SWAPI data products from Level 1 through Level 3.

Level 1 Data Products

- **Counts and Count Uncertainties:**
 - **Time resolution:** 12 s (72 values per sweep, coarse + fine sweep)
 - **Components:** primary, secondary, coincidence
 - **Inputs Required:** Level 0 CCSDS packets
 - **Ancillary files:** Decommuation Table

Level 2 Data Products

- **Count Rates and Count Rate Uncertainties:**
 - **Time resolution:** 12 s
 - **Components:** primary, secondary, coincidence
 - **Inputs Required:** Level 1 counts and count uncertainties as a function of 72 ESA voltage steps (per sweep)
 - **Ancillary files:** ESA unit conversion table, spacecraft thruster data

Level 3a Data Products

1. Solar Wind Proton (H^+) Products:

- **Velocity ($km\ s^{-1}$)**
 - **Time resolution:** 1 min
 - **Components:** (v_{SW} , θ , ϕ), IMAP de-spun frame
 - **Inputs Required:** Level 2 count rates and count rate uncertainties as a function of E/q for 62 coarse energy bins (5 sweeps)
 - **Ancillary files:** IMAP spin phase (SPICE kernels), LUT for the speed and flow angles
- **Density (cm^{-3})**
 - **Time resolution:** 1 min
 - **Components:** n_{SW}

- o **Inputs Required:** Level 2 count rates and count rate uncertainties as a function of E/q for 62 coarse energy bins (5 sweeps)
 - o **Ancillary files:** Effective area (A_{eff}), FWHM of energy width ($\frac{\Delta E}{E}$), LUT for SW proton density and temperature
 - **Temperature (K)**
 - o **Time resolution:** 1 min
 - o **Components:** T_{SW} (isotropic)
 - o **Inputs Required:** Level 2 count rates and count rate uncertainties as a function of E/q for 62 coarse energy bins (5 sweeps)
 - o **Ancillary files:** same as for density
2. **Solar Wind Alpha ($^4\text{He}^{++}$) Products:**
- **Speed (km s^{-1})**
 - o **Time resolution:** 1 min
 - o **Components:** $v_{\text{He}^{++}}$
 - o **Inputs Required:** Level 2 count rates and count rate uncertainties as a function of E/q for 62 coarse energy bins (5 sweeps)
 - o **Ancillary files:** NA
 - **Density (cm^{-3})**
 - o **Time resolution:** 1 min
 - o **Components:** $n_{\text{He}^{++}}$
 - o **Inputs Required:** Level 2 count rates and count rate uncertainties as a function of E/q for 62 coarse energy bins (5 sweeps)
 - o **Ancillary files:** Effective area (A_{eff}), FWHM of energy width ($\frac{\Delta E}{E}$), LUT for SW alpha density and temperature
 - **Temperature (K)**
 - o **Time resolution:** 1 min
 - o **Components:** $T_{\text{He}^{++}}$ (isotropic)
 - o **Inputs Required:** Level 2 count rates and count rate uncertainties as a function of E/q for 62 coarse energy bins (5 sweeps)
 - o **Ancillary files:** Same as for density
3. **Pickup Helium (He^+) Products:**
- **Fit Parameters:** Cooling index, Cutoff speed (km s^{-1}), Ionization Rate (s^{-1}), Background Rate (Hz)
 - o **Time resolution:** 10 min
 - o **Components:** α , v_b , β_E , C_{bkg}
 - o **Inputs Required:** Level 2 count rates and count rate uncertainties as a function of E/q for 62 coarse energy bins (50 sweeps)
 - o **Ancillary files:** IMAP spin phase (SPICE), Frame kernel from SWAPI to GSE (SPICE), Frame kernel from GSE to Ecliptic J2000 (SPICE), LUT for the energy-dependent geometric factor, LUT for the efficiency, LUT for the instrument response function, LUT for the interstellar neutral He density
 - **Density (cm^{-3})**
 - o **Time resolution:** 10 min
 - o **Components:** n_{He^+}
 - o **Inputs Required:** Level 2 count rates and count rate uncertainties as a function of E/q for 62 coarse energy bins (50 sweeps)
 - o **Ancillary files:** Same as for fit parameters

- **Temperature (K)**

- o **Time resolution:** 10 min
- o **Components:** T_{He+} (isotropic)
- o **Inputs Required:** Level 2 count rates and count rate uncertainties as a function of E/q for 62 coarse energy bins (50 sweeps)
- o **Ancillary files:** Same as for fit parameters

Level 3b Data Products

- **Combined differential intensity**

- o **Time resolution:** 10 min
- o **Components:** $J(E/q)$
- o **Inputs Required:** Level 2 count rates and count rate uncertainties as a function of E/q for 62 coarse energy bins (50 sweeps)
- o **Ancillary files:** $\frac{\Delta E}{E}$, energy-dependent geometric factor ($G(E/q)$), efficiency (ϵ)

Acknowledgements We gratefully acknowledge all the scientists, engineers, technicians, administrators, and other personnel who have made the SWAPI instrument and the IMAP mission a reality. SWAPI was the first NASA flight instrument to be developed, assembled, & tested by the Space Physics Group at Princeton. Success of this effort reflects the dedication and collaboration of individuals not only at Princeton, but across multiple institutions, each contributing critical expertise in design, manufacturing, assembly, testing, operations, documentation, and analysis. Although it is not possible to name everyone who has contributed, we wish to recognize the outstanding efforts of: Glen Atkinson, Jeffrey Boye, Mathew Bunday, Jianxin Chen, Michael DeLuca, Eric Diamonte, Brian Fennimore, Burt Harrop, Rachel Hartig, Jeff Leese, Steve Lowe, Julie Luster, Jackie Perry, Susan Pope, Homer Rodriguez, Lydia Rojek, Kristen Seith, Teji Sharma, Ken Smith, Carlos Urdiales, and Dan White, whose work, alongside that of the rest of the team, has been instrumental in bringing SWAPI from concept to flight. We also gratefully acknowledge the contributions of our students whose diligence and dedication were instrumental in the development of the Space Physics Laboratory's Infrastructure and supporting the various phases of SWAPI's calibration: Ariane Adcroft, Abhinav Agarwal, Nick Allen, Nina Arcot, Gina Arnau, Jian Arnold, Joey Avallone, Sonal Bhatia, Mila Bileska, Wolf Cukier, Jupiter Ding, Candace Do, Mark Dodici, Tyler Eddy, Sydney Evans, Shannon Filer, Osvaldo Flores, Aditya Gandotra, Grace Gong, Patrick Grate, Teodor Grosu, Zakiya Helm, Michael Hwang, Rishabh Jain, William Xander Jenkin, Inci Karaalan, Isabel Kim, Bethwel Kiplimo, Elif Kozanglu, Veena Krishnaraj, William Li, Raina Maldonado, Anthony Matimu, Cole Meyer, Andres Montoya, Mori Ono, Jahnvi Padukone, Alexandros Papamathaiou, Daniel Pardo, Savannah Pobre, Rishika Porandla, Joseph Radel, Ariana Rausch, Marie Riddle, Christian Robles, Sam Ruona, Mariel Shapiro, Kate Sheldon, Rala Abu Sulami, Levon Tabirian, Kazuki Tojo, and Susan Zhang. SWAPI and the rest of IMAP would not have been possible without the dedication of many people, and we are honored to recognize even a portion of the collective effort that made it a success. Some of this work was performed in part at the Micro/Nano Fabrication Center (MNFC), a core shared-use facility of the Princeton Materials Institute (PMI). This work is supported by the IMAP mission as a part of NASA's STP mission line (80GSFC19C0027).

Declarations

Competing Interests The authors have no conflicts of interest to declare that are relevant to the content of this article.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

References

- Agostinelli S, Allison J, Amako K, Apostolakis J, Araujo H, Arce P, et al (2003) Geant4—a simulation toolkit. *Nucl Instrum Methods Phys Res, Sect A* 506(3):250–303. [https://doi.org/10.1016/S0168-9002\(03\)01368-8](https://doi.org/10.1016/S0168-9002(03)01368-8)
- Allison J, Amako K, Apostolakis J, et al (2006) Geant4 developments and applications. *IEEE Trans Nucl Sci* 53:270–278. <https://doi.org/10.1109/TNS.2006.869826>
- Allison J, Amako K, Apostolakis J, et al (2016) Recent developments in Geant4. *Nucl Instrum Methods Phys Res, Sect A, Accel Spectrom Detect Assoc Equip* 835:186–225. <https://doi.org/10.1016/j.nima.2016.06.125>
- Baker S, Cousins RD (1984) Clarification of the use of CHI-square and likelihood functions in fits to histograms. *Nucl Instrum Methods Phys Res* 221:437–442
- Bzowski M, Wawrzaszek R, Strumik M, et al. (2025). GLObal solar Wind Structure (GLOWS). *Space Sci Rev* 221. <https://doi.org/10.1007/s11214-025-01233-y>
- Cash W (1979) Parameter estimation in astronomy through application of the likelihood ratio. *Astrophys J* 228:939–947
- Chen JH, Bochsler P, Möbius E, Gloeckler G (2014) Possible modification of the cooling index of interstellar helium pickup ions by electron impact ionization in the inner heliosphere. *J Geophys Res Space Phys* 119:7142–7150. <https://doi.org/10.1002/2014JA020357>
- Cohen CMS, Alterman BL, Baker DN (2025) IMAP's role in understanding particle injection and energization throughout the heliosphere. Submitted to *Space Sci Rev*
- Dahl DA (2000) SIMION for the personal computer in reflection. *Int J Mass Spectrom* 200:3–25. [https://doi.org/10.1016/S1387-3806\(00\)00305-5](https://doi.org/10.1016/S1387-3806(00)00305-5)
- Ebert RW, McComas DJ, Rodriguez B, Valek P, Weidner S (2010) A composition analysis tool for the Solar Wind Around Pluto (SWAP) instrument on new horizons. *Space Sci Rev* 156:1–12. <https://doi.org/10.1007/s11214-010-9683-6>
- Elliott HA, McComas DJ, Valek P, Nicolaou G, Weidner S, Livadiotis G (2016) The new horizons Solar Wind Around Pluto (SWAP) observations of the solar wind from 11–33 au. *Astrophys J Suppl Ser* 223:19. <https://doi.org/10.3847/0067-0049/223/2/19>
- Funsten HO, Harper RW, McComas DJ (2005) Absolute detection efficiency of space-based ion mass spectrometers and neutral atom imagers. *Rev Sci Instrum* 76:053301. <https://doi.org/10.1063/1.1889465>
- Gloeckler G, Geiss J (1998) Interstellar and inner source pickup ions observed with SWICS on ULYSSES. *Space Sci Rev* 86:127–159. <https://doi.org/10.1023/A:1005019628054>
- Hegarty KP, Cully M, Kubota S, et al (2025) Observatory Overview. Submitted to *Space Sci Rev*
- Ji X, Gu W, Qian X, Wei H, Zhang C (2020) Combined Neyman–Pearson chi-square: an improved approximation to the Poisson-likelihood chi-square. *Nucl Instrum Methods Phys Res, Sect A, Accel Spectrom Detect Assoc Equip* 961:163677
- Khoo LY, McComas DJ, Rankin JS, Shen MM, Sharma T, Shi C (2023) Compensating for gyroradius effects in beamlines with small Helmholtz coils. *Rev Sci Instrum* 94:035102. <https://doi.org/10.1063/5.0135154>
- King JH, Papitashvili NE (2005) Solar wind spatial scales in and comparisons of hourly wind and ACE plasma and magnetic field data. *J Geophys Res Space Phys*. <https://doi.org/10.1029/2004JA010649>
- Lallement R, Quémerais E, Bertaux JL, Ferron S, Koutroumpa D, Pellinen R (2005) Deflection of the interstellar neutral hydrogen flow across the heliospheric interface. *Science* 307:1447–1449. <https://doi.org/10.1126/science.1107953>
- Lee CO, Christian ER, Sandoval L, et al (2025) Space Weather Science to Enhance Forecasting with the NASA IMAP Active Link for Real-Time (I-ALiRT) System. Submitted to *Space Sci Rev*
- Lindsay BG, Stebbings RF (2005) Charge transfer cross sections for energetic neutral atom data analysis. *J Geophys Res Space Phys*. <https://doi.org/10.1029/2005JA011298>
- Livi S, Desai MI, Ogasawara K, et al (2025) For the Compact Dual Ion Composition Experiment: CoDICE. Submitted to *Space Sci Rev*
- McComas DJ, Bame SJ (1984) Channel multiplier compatible materials and lifetime tests. *Rev Sci Instrum* 55:463–467. <https://doi.org/10.1063/1.1137797>
- McComas DJ, Baldonado JR, Bame SJ, Barraclough BL (1987) Channel electron multiplier compatibility with Viton and Apiezon-L vacuum grease. *Rev Sci Instrum* 58:2331–2332. <https://doi.org/10.1063/1.1139346>
- McComas DJ, Allegrini F, Pollock CJ, Funsten HO, Ritzau S, Gloeckler G (2004) Ultrathin (~10 nm) carbon foils in space instrumentation. *Rev Sci Instrum* 75:4863–4870
- McComas DJ, Allegrini F, Bagenal F, et al (2008) The Solar Wind Around Pluto (SWAP) instrument aboard New Horizons. *Space Sci Rev* 140:261–313. <https://doi.org/10.1007/s11214-007-9205-3>

- McComas DJ, Bzowski M, Fuselier SA, et al (2015) Local interstellar medium: six years of direct sampling by IBEX. *Astrophys J Suppl Ser* 220:22. <https://doi.org/10.1088/0067-0049/220/2/22>
- McComas DJ, Zirnstein EJ, Bzowski M, et al (2017) Interstellar pickup ion observations to 38 au. *Astrophys J Suppl Ser* 233:8. <https://doi.org/10.3847/1538-4365/aa91d2>
- McComas DJ, Christian ER, Schwadron NA, et al (2018) Interstellar Mapping and Acceleration Probe (IMAP): a new NASA mission. *Space Sci Rev* 214:116. <https://doi.org/10.1007/s11214-018-0550-1>
- McComas DJ, Swaczyna P, Szalay JR, Zirnstein EJ, Rankin JS, Elliott HA, Singer K, Spencer J, Stern SA, Weaver H (2021) Interstellar pickup ion observations halfway to the termination shock. *Astrophys J Suppl Ser* 254:19. <https://doi.org/10.3847/1538-4365/abee76>
- McComas DJ, Shrestha BL, Swaczyna P, et al (2022) First high-resolution observations of interstellar pickup ion mediated shocks in the outer heliosphere. *Astrophys J* 934:147. <https://doi.org/10.3847/1538-4357/ac7956>
- McComas DJ, Christian ER, Schwadron NA, et al (2025b) Interstellar Mapping and Acceleration Probe: the NASA IMAP mission. *Space Sci Rev* 221:100. <https://doi.org/10.1007/s11214-025-01224-z>
- McComas DJ, Shrestha BL, Livadiotis G, et al (2025a) Interstellar pickup ion observations to 60 au. *Astrophys J* 980:154. <https://doi.org/10.3847/1538-4357/ada6b1>
- Nicolaou G, McComas DJ, Bagenal F, Elliott HA (2014) Properties of plasma ions in the distant Jovian magnetosheath using Solar Wind Around Pluto data on New Horizons. *J Geophys Res Space Phys* 119:3463–3479. <https://doi.org/10.1002/2013JA019665>
- Piessens R, De Doncker-Kapenga E, Überhuber CW, Kahaner DK (1983) Quadpack: a subroutine package for automatic integration. Springer Series in Computational Mathematics, vol 1. Springer, Berlin. <https://doi.org/10.1007/978-3-642-61786-7>
- Randol BM, Ebert RW, Allegrini F, McComas DJ, Schwadron NA (2010) Reflections of ions in electrostatic analyzers: a case study with New Horizons/Solar Wind Around Pluto. *Rev Sci Instrum* 81:114501. <https://doi.org/10.1063/1.3499367>
- Reisenfeld DB, Allegrini FA, Christian ER, et al (2025) Exploring the outer heliosphere through ENA Observations from IMAP. Submitted to *Space Sci Rev*
- Reno C, Lafferty G, Lucas G, Ord K, et al (2025) IMAP Science and Mission Operations. Submitted to *Space Sci Rev*
- Schwadron NA (1998) A model for pickup ion transport in the heliosphere in the limit of uniform hemispheric distributions. *J Geophys Res* 103:20643–20650
- Schwadron NA, Geiss J, Fisk LA, Gloeckler G, von Zurbuchen TH, Steiger R (2000) Inner source distributions: theoretical interpretation, implications, and evidence for inner source protons. *J Geophys Res* 105:7465–7472
- Schwadron N, Shen M, Begley S, et al. (2025) The IMAP-Lo instrument. *Space Sci Rev* 221. <https://doi.org/10.1007/s11214-025-01233-y>
- Shrestha BL, Zirnstein EJ, McComas DJ, Brandt P, Stern A, Elliott HA, Poppe AR, Singer KN, Verbiscer A (2024) Suprathermal H⁺ pickup ion tails in the outer heliosphere. *Astrophys J* 960:35. <https://doi.org/10.3847/1538-4357/ad08b9>
- Stern SA (2008) The New Horizons Pluto Kuiper Belt mission: an overview with historical context. *Space Sci Rev* 140:3–21. <https://doi.org/10.1007/s11214-007-9295-y>
- Szalay JR, Provornikova E, Ayari E, et al (2025) Direct Samples of Interstellar and Interplanetary Material: Expectations for IMAP. Submitted to *Space Sci Rev*
- Thomas GE (1978) The interstellar wind and its influence of the interplanetary environment. *Annu Rev Earth Planet Sci* 6:173. <https://doi.org/10.1146/annurev.ea.06.050178.001133>
- Vasyliunas VM, Siscoe GL (1976) On the flux and the energy spectrum of interstellar ions in the Solar System. *J Geophys Res* 81:1247–1252. <https://doi.org/10.1029/JA081i007p01247>
- Zirnstein EJ, McComas DJ, Kumar R, Elliott HA, Szalay JR, Olkin CB, Spencer J, Stern SA, Young LA (2018) In situ observations of preferential pickup ion heating at an interplanetary shock. *Phys Rev Lett* 121:075102. <https://doi.org/10.1103/PhysRevLett.121.075102>
- Zirnstein EJ, Möbius E, Zhang M, Bower J, Elliott HA, McComas DJ, Pogorelov NV, Swaczyna P (2022) In situ observations of interstellar pickup ions from 1 au to the outer heliosphere. *Space Sci Rev* 218:28. <https://doi.org/10.1007/s11214-022-00895-2>